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Methodological basis for quantitative reconstruction of air temperature and sunshine from pollen assemblages in Arctic Canada and Greenland

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ABSTRACT

This study presents a modern database including 831 pollen assemblages from lakes of the Boreal, Subarctic and Arctic biomes of North America and Greenland, and corresponding temperature, sunshine and precipitation. Pollen data include the 39 most common vascular taxa (18 woody plants and 21 herbs). Multivariate ordinations using correspondence analysis (CA) and canonical correspondence analysis (CCA) reveal that temperature and sunshine account, respectively, for 45.5% and 44.4% of the variance within pollen assemblages, whereas precipitation only accounts for 10.2% of total variance. CCA further demonstrates that the climatic information encapsulated in pollen assemblages is not the same for the three biomes. In the Boreal biome, precipitation and temperature account for most of the variance, whereas sunshine and precipitation are more determinant in the Subarctic biome, and the temperature and sunshine seem to exert the main control on the pollen distribution in the Arctic biome. The modern analogue technique (MAT) and CCA regressions were tested for reconstructing climate parameters. MAT yields better results than CCA regressions and validation tests indicate a root mean square error (RMSE) of 1.35 °C for July air temperature, 2.3% for July sunshine, 3.10 °C for January air temperature and 143 mm for annual precipitation. Two examples of reconstructions are presented from recent (<450 years) lake-sediment cores of eastern Baffin Island, Arctic Canada and southwest Greenland. They demonstrate that July air temperature and July sunshine can be reconstructed independently from Arctic pollen assemblages, but January air temperature and annual precipitation must be interpreted with caution. They also suggest that reconstruction of past sunshine variations can serve to document climate changes at synoptic scale including the North Atlantic Oscillation (NAO).

1. Introduction

In arctic environments, pollen analysis may provide valuable information about past vegetation and climate changes (e.g., Ritchie, 1984, 1987; Gajewski and MacDonald, 2004; Gajewski, 2006). The Holocene climate history of the western Canadian Arctic and Alaska (Anderson et al., 1989; Bigelow et al., 2003; Kaplan et al., 2003; Zabenskie and Gajewski, 2007; Peros and Gajewski, 2008), Scandinavia (Seppä and Birks, 2001, 2002) and northern Eurasia (Andreev et al., 2003, 2004a–c, 2005) has been documented based on pollen analyses. However, quantitative climate reconstructions from pollen records are still rare in the eastern Canadian Arctic and Greenland (Kerwin et al., 2004; Fréchette et al., 2006).

Unlike the boreal forest, arctic vegetation is typically characterized by high diversity of species and pronounced heterogeneity at local to regional scales. Therefore, the pollen rain registered in arctic lake sediments may vary considerably from one site to another, even at sites located in close proximity and experiencing similar climate (Kerwin, 2000; Gajewski, 2002; Fréchette et al., 2006). Therefore, the understanding of the relationship between recent high-latitude pollen assemblages and modern climate is necessary before attempting to reconstruct past climate from arctic pollen assemblages. However, in arctic

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environments, comprehensive assessment of modern pollen vs. climate relationship is still pending. Thus, we have compiled a modern database, which includes pollen data counts in 831 surface sediment samples from lakes of the Boreal-Arctic North America and Greenland (Fig. 1A). By focusing on lake sediments, and discarding peat, wetland or soil samples, we aim at limiting the effect of over-representation of the local vegetation in pollen assemblages as it may cause heterogeneity independent from large-scale climate control. This approach is further guided by the general suitability and great promise of lake sediments for the study of high-latitude environmental change (e.g., Pienitz et al., 2004). The modern climate parameters considered in this study include temperature (T), precipitation (P) and sunshine (S). The analyses were made by considering three biomes (see Figs. 1A and 2), which are major, distinctive complex of ecologically similar communities of plants, animals, and soil organisms and have a distribution controlled by climate. The distribution of the Boreal, Subarctic and Arctic biomes of North America and Greenland is given in Fig. 2. The Boreal biome holds the boreal forest. The Arctic biome includes the rock desert and the High, Mid and Low Arctic tundra vegetation types. The Subarctic biome corresponds to the transition between the Boreal and the Arctic biomes and occurs across the roughly 10 °C mean July air temperature gradient (e.g., Bonan et al., 1992; Foley et al., 1994; Pielke and Vidale, 1995; Walker et al., 2002). It includes the subarctic tundra and the forest tundra vegetation types (see Fig. 2 legend for more details).

One of the originality of this paper is the use of sunshine, which is inversely proportional to cloud cover and plays an important role in pollen production (cf. Rodríguez-Rajo et al., 2005). Changes in cloudiness are associated with changes in cyclone frequency (Serreze et al., 2000). The reconstruction of past cloud cover variability can thus be helpful to understand the mechanisms behind shifts in cyclone tracks (cf. Previdi and Veron, 2007; Tsukernik et al., 2007) and then to document long-term changes in synoptic scale atmospheric circulation patterns, like the North Atlantic Oscillation (NAO) (Hurrell, 1995).

Our main objectives are (1) to document the relationship between modern pollen assemblages and climate parameters in the Boreal, Subarctic and Arctic biomes, (2) to develop methods for the reconstruction of climate parameters in the Canadian Arctic and Greenland after standardization and harmonization of pollen taxonomy, (3) to evaluate the accuracy of the methods based on reconstructions in two recent (<450 years) sedimentary sequences from lakes of eastern Baffin Island and southwest Greenland.

2. Modern database

2.1. Pollen data

The modern database of pollen assemblages and climate parameters used in this study was developed from the modern

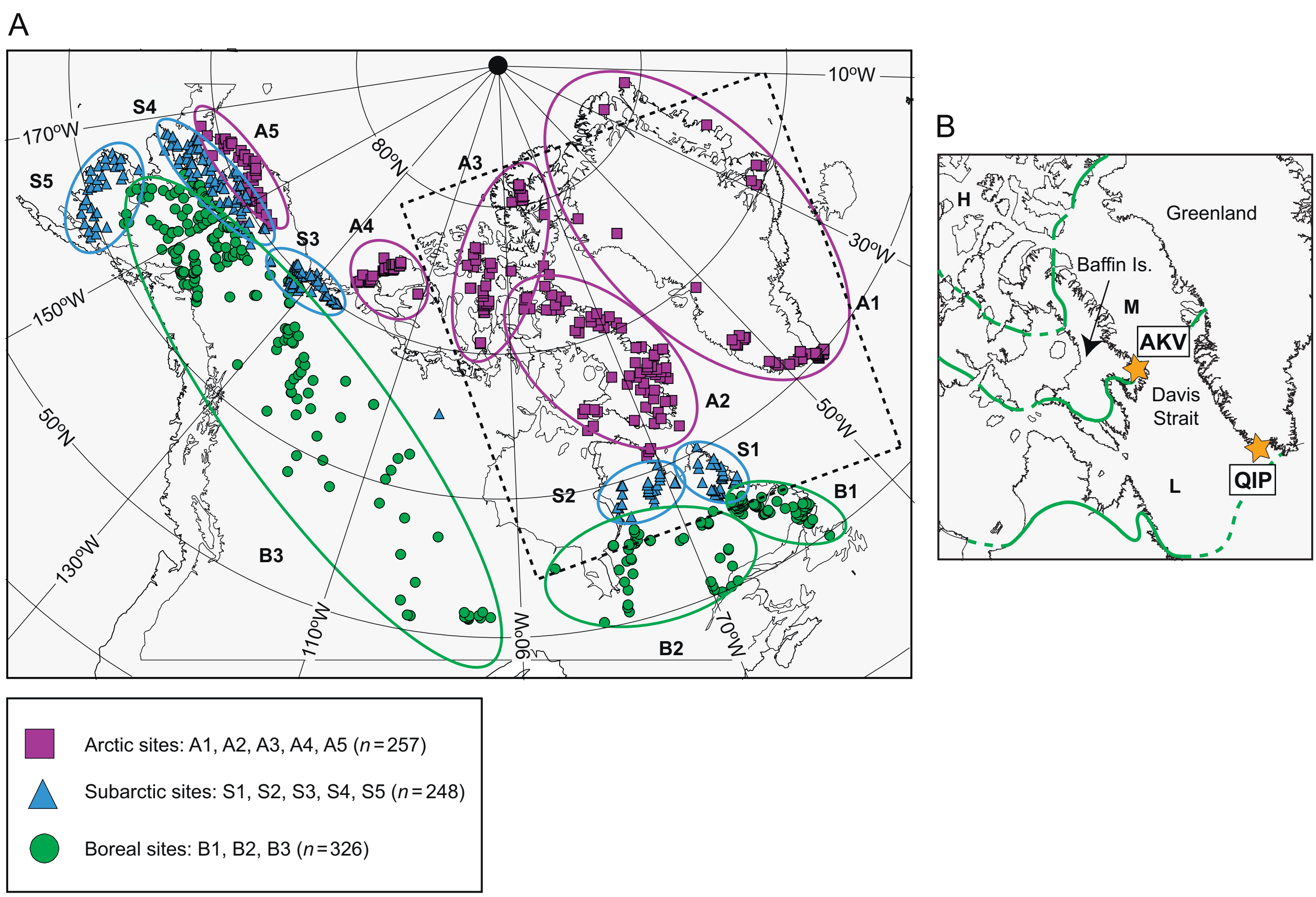

Fig. 1. (A) Locations of the 831 lake sediment samples used to develop the modern pollen database. The sites have been divided by biome (A: Arctic, S: Subarctic, B: Boreal) in which they are grouped according to their geographical location. The square delimitated by a dotted line corresponds to the area in (B). (B) Location map of Akvaqiak Lake (66°47′N, 63°57′W, 45 m a.s.l.) on northern Cumberland Peninsula, Baffin Island, and Qipisarqo Lake (61°00′N, 47°45′W, 7 m a.s.l.) on southwest Greenland. The green lines show approximate limits of low (L), mid (M) and high (H) Arctic vegetation zones after Young (1971).

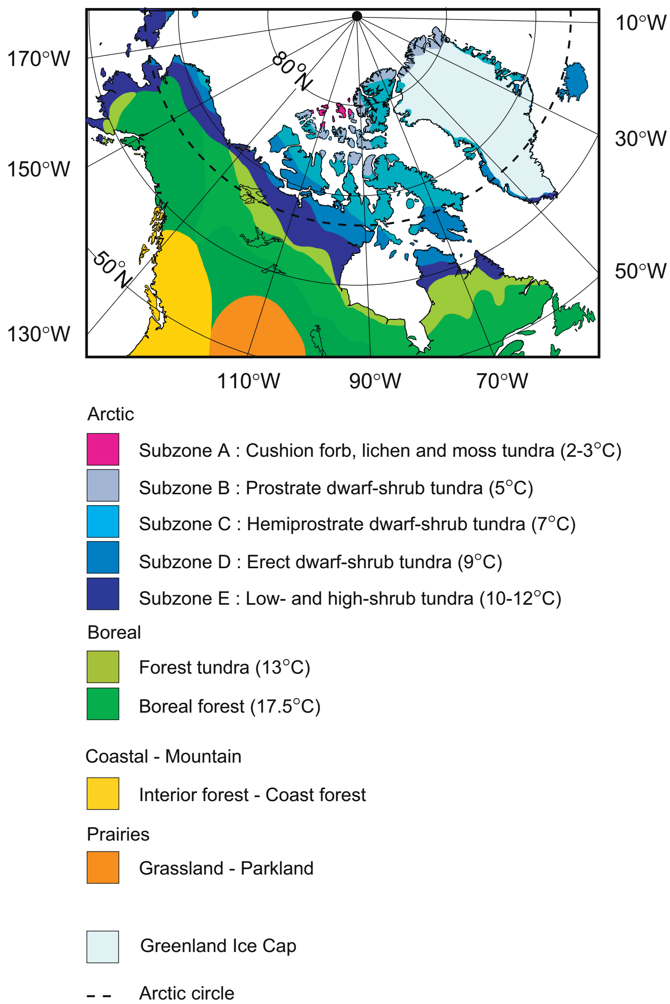

Fig. 2. Main vegetation zones (or biomes) of northern North America and Greenland as synthesized from Young (1971), Payette (1983), Anderson et al. (1989), Fedorova et al. (1990), Gould et al. (2002), Walker (2000), Walker et al. (2002), CAVM Team (2003) and Dyke (2005). A complete description of the five subzones of the Arctic biome is available in Walker et al. (2002). The Rock desert is equivalent to the Subzone A, the High Arctic tundra to the Subzone B, the Mid-Arctic tundra to the Subzone C, the Low Arctic tundra to the Subzone D and the Subarctic tundra to the Subzone E. The mean July air temperature at the southern boundary of the Arctic and Boreal biomes is given. Those for the Arctic biome are from Walker et al. (2002), the one for the forest tundra is from Edlund (1986) and the one for the boreal forest is from Hare and Thomas (1974). According to the 10 °C July air temperature isotherm we included the low- and high-shrub tundra and the forest tundra in the Subarctic biome.

pollen network compiled by Whitmore et al. (2005) for North America and Greenland. The data files and associated metadata are available on the web pages of the National Geophysical Data Center (www.ngdc.noaa.gov/paleo), the Laboratory for Palaeoclimatology and Climatology (www.lpc.uottawa.ca), and the University of Wisconsin Geography Department (http://www.geography.wisc.edu/faculty/williams/data/data.html).

From the 4634 samples of this modern pollen network, we first selected sites north of latitude 50° ($n = 1615$). We then retained sites from lake environment ($n = 1147$) in which the pollen frequency is expressed in raw counts reaching a sum of at least 100 grains ($n = 917$). Two spectra from surface lake-sediment samples (eastern Baffin Island, southwestern Greenland) were added. Each of the 919 sites was assigned to a biome based on comparison with distribution map of major vegetation zones (Gould et al., 2002; Walker et al., 2002; CAVM Team, 2003; Kaplan et al., 2003) (Fig. 2) and their International Geosphere–Biosphere Programme (IGBP) and Global Ecosystems (GLEC) classification schemes (Whitmore et al., 2005). Sites from Coastal Mountain ($n = 50$ sites) and Prairie ($n = 38$ sites) biomes were discarded. The final database includes 831 surface lake-sediment pollen assemblages and corresponding climate parameters from the Boreal ($n = 326$), Subarctic ($n = 248$) and Arctic ($n = 257$) biomes (Fig. 1A).

The 831 pollen frequency data matrix contains 94 spermatophyte taxa. As pollen counts were performed in various laboratories by different analysts, problems of inconsistent taxonomy and nomenclature may arise (Birks, 1995). Therefore, simplification and rationalization of the data matrix was made following several lines of ecological and statistical reasoning, as summarized below.

In arctic lakes, wind-blown exotic pollen grains produced by boreal forest trees and shrubs often occur in large number. Their proportions are function of the distance from the source vegetation, wind direction and speed during pollination, and the dilution by pollen produced locally and regionally by herbs and shrubs (von Post, 1916 in Davis and Faegri, 1967; Bourgeois et al., 2001; Gajewski, 2002). A strategy to avoid bias due to the over-representation of long-distance input is to omit wind-blown exotic taxa from the pollen sum. However, such a practice remains problematic because percentages of local taxa increase when the assumed long-distance component is excluded. Moreover, any climatic information pertaining to southerly atmospheric flow provided by exotic pollen transport is lost (Calcote, 2003; Jackson and Williams, 2004; Fréchette et al., 2006). Therefore, some taxa exotic to the Boreal, Subarctic and Arctic biomes, such as *Ulmus*, *Fagus*, *Quercus*, *Fraxinus*, Polemoniaceae, Chenopodiaceae and *Ambrosia* were nevertheless kept in the database (see Table 1).

The inclusion of pteridophyte spores was considered. However, the identification of pteridophyte spores as monolete or trilete spores without any taxonomic indication by some authors (e.g., Richard, 1981; Kerwin, 2000) and the absence of pteridophyte in some spectra (e.g., Davis and Webb, 1975) precluded the inclusion of pteridophyte taxa in the sum.

Further taxonomic concern was raised with respect to *Picea*, *Pinus*, and *Alnus*, which were rarely identified at the species level. Similarly, the differentiation of Ericaceae tetrads at the genus level was only occasionally reported. This limitation also applies to many herb taxa. Therefore, among the 94 pollen taxa counted and reported by the different authors, we reclassified several taxa to higher taxonomic level in order to ensure consistent taxononomical nomenclature as recommended by Birks (1995). A total of 64 taxa remained after this harmonization. Finally, to reduce the statistical noise induced by rare species, we excluded all taxa occurring less than twice with a minimum relative frequency of 1% in the 831 samples. After this exercise, the pollen database is composed of 39 spermatophyte taxa (cf. Table 1; information on taxa grouping is given on Supplementary Table S1). In the overall database (831×39 matrix), the number of taxa per sample varies from 3 to 25, with a mean of 14 ± 4, and the sum used for calculation of relative frequencies ranges between 102 and 3357 grains per sample, with a mean of 461 ± 239. The pollen assemblages of the 831 modern samples are portrayed in Fig. 3 as relative frequencies of the 39 principal taxa ordered according to their first axis canonical correspondence analysis (CCA) scores constrained with July air temperature only (cf. Section 3.2.3) and ordered by latitude within each region.

2.2. Climate data

We compiled the elevation-corrected climatic data from the data set of Whitmore et al. (2005). The climatic data were

Table 1
List of the 39 pollen taxa retained in the modern database that includes 831 sites from surface lake sediment

Taxa	Code	Range of %	Mean % ± SD	*N*
Woody plants: trees+shrubs				
*Ulmus**	ULMA	0–1.99	0.02 ± 0.10	35
*Fagus**	FFAG	0–1.02	0.01 ± 0.06	11
*Quercus**	FQUE	0–2.34	0.05 ± 0.20	93
Abies	PABI	0–9.55	0.23 ± 0.89	119
Myrica	MMYR	0–4.46	0.11 ± 0.36	168
Populus	SPOP	0–4.13	0.14 ± 0.40	186
Larix	PLAR	0–2.57	0.12 ± 0.31	181
Corylus	BCOR	0–1.96	0.02 ± 0.15	43
*Fraxinus**	OFRA	0–1.02	0.02 ± 0.09	55
Picea	PPIC	0–90.98	22.02 ± 22.55	794
Eleagnaceae	ELAE	0–1.54	0.02 ± 0.10	45
Pinus	PPIN	0–64.91	4.11 ± 9.00	554
Alnus	BALN	0–76.62	16.78 ± 13.38	820
Betula	BBET	0–71.20	20.80 ± 12.53	830
Cupressaceae	CUPR	0–2.86	0.11 ± 0.30	173
Ericales	ERIC	0–50.48	3.94 ± 6.83	700
Salix	SSAL	0–50.46	3.62 ± 5.91	737
Dryas	RDRY	0–4.79	0.13 ± 0.43	131
Herbs				
Polemoniaceae*	POLE	0–5.10	0.01 ± 0.18	22
Apiaceae	APIA	0–1.69	0.02 ± 0.12	48
Chenopodiaceae*	CHEN	0–4.12	0.13 ± 0.35	200
*Ambrosia**	CAMB	0–15.09	0.41 ± 0.94	330
Tubuliflorae/Liguliflorae	TULI	0–4.03	0.34 ± 0.55	448
Onagraceae	ONAG	0–1.09	0.02 ± 0.10	67
Artemisia	CART	0–13.77	1.06 ± 1.54	630
Poaceae	POAC	0–89.02	7.03 ± 9.17	780
Thalictrum	RTHA	0–11.16	0.08 ± 0.53	89
Rosaceae	ROSA	0–11.11	0.50 ± 0.89	480
Cyperaceae	CYPE	0–89.38	15.35 ± 14.72	812
Ranunculaceae	RANU	0–12.57	0.25 ± 0.90	281
Fabaceae	FABA	0–5.17	0.07 ± 0.26	148
Polygonaceae	POLY	0–3.00	0.04 ± 0.18	84
Plantago	PPLA	0–1.32	0.02 ± 0.11	53
Caryophyllaceae	CARY	0–5.75	0.23 ± 0.55	270
Scrophulariaceae	SCRO	0–2.03	0.04 ± 0.17	68
Brassicaceae	BRAS	0–11.18	0.26 ± 0.86	192
Oxyria/Rumex	POXR	0–35.63	1.24 ± 3.66	305
Saxifragaceae	SAXI	0–13.86	0.59 ± 1.68	239
Papaveraceae	PPAV	0–7.74	0.08 ± 0.51	56

See Supplementary Table S1 for information on taxa grouping, and summary statistics on the occurrence of these taxa. Taxa are ordered according to their scores on the primary CCA axis constrained with July air temperature only. Asterisks (*) indicate taxa that do not presently occur in the Boreal, Subarctic and Arctic biomes of North America and Greenland.

calculated from the Climatic Research Unit (CRU) gridded climatology using 1961–1990 climatological averages (New et al., 2002) and interpolated to each pollen sites via lapse-rate collected from bilinear interpolation. Whitmore et al. (2005) describes in detail the method used to estimate modern climate. Herein, we developed a matrix with mean monthly temperature, precipitation and sunshine data, average annual temperature and total annual precipitation. The gridded sunshine data provided by New et al. (2002) and reported by Whitmore et al. (2005) are expressed as percent of possible bright sunshine and were derived from sunshine averages at given stations or from conversion of cloud cover to sunshine. The accuracy of the monthly gridded climatology varies both regionally and between variables (New et al., 1999). The squared root generalized cross-validation (RTGCV) statistics provide an estimate of the mean predictive error (and hence power) of interpolated variables in the study area (see New et al. (1999, 2002) for the RTGCV of mean monthly temperature, precipitation and sunshine data in North America and Greenland).

3. Numerical methods

3.1. Ordination

The relationship between pollen assemblages and climate parameters (mean monthly temperature, precipitation and sunshine) was explored statistically. The relationship between the relative pollen abundance (e.g., *Picea* %) and climate variables is provided elsewhere (Williams et al., 2006). Detrended correspondence analysis (DCA, with detrending by segments) was first used to estimate the compositional gradient lengths of the first few DCA axes in the modern pollen data (Gauch, 1982; ter Braak and Prentice, 1988; Birks, 1995). The gradient length of DCA axis 1 is 2.80 standard deviation units, which indicates unimodal response of pollen vs. climatic gradients and suggests that correspondence analysis (CA) and canonical correspondence analysis (CCA) provide suitable approaches. CA was used to detect major floristic gradients within the 831 modern pollen assemblages, and CCA was thereafter applied to evaluate how climatic data explain the biotic response included in the pollen assemblages. CCAs were done with all biomes collectively and with each biome individually. Two climatic matrices were used. The first one includes 38 climatic parameters (all mean monthly temperature, precipitation and sunshine data, and mean annual temperature and precipitation) and enables a first-order appreciation of the climate variables that maximize the floristic gradients identified by CA. The second one includes only eight climatic parameters (mid-summer and winter monthly temperature, precipitation and sunshine, and annual temperature and precipitation) and allows the study of the relationship between pollen taxa and climate variables.

Forward-selection procedures in CCA (*sensu* ter Braak and Verdonschot, 1995) were applied to assess the explanatory power of climatic variables. The climatic variables were ranked by importance on the basis of their marginal and conditional effects (*sensu* ter Braak and Verdonschot, 1995). The first series of CCAs used each climatic variable as the sole constraining variable (marginal effect). After selection of the best climatic variable, all remaining variables are ranked through a further series of CCAs with each one being alternately chosen as the sole additional variable beyond those already selected (conditional effect). The statistical significance of each variable was established with 999 unrestricted Monte-Carlo permutations with the application of a Bonferroni correction to the reported p-values (Legendre and Legendre, 1998; Birks et al., 2004).

The CCA results were also compared to those of the unconstrained (CA) ordination in order to verify if the climatic parameters used in the CCA retain the major floristic gradients suggested from CA in the absence of *a priori* climatic constraints. For DCA, CA, and CCA, the frequencies of the 39 pollen taxa were first square-root transformed in order to reduce the weight of the most abundant taxa and to increase the weight of accompanying taxa. This transformation is necessary because many arctic plants, notably herbaceous taxa, are sytematically under-represented due to low pollen production, whereas the pollen of shrubs such as *Alnus* and *Betula* is ubiquitous in the Boreal, Subarctic and Arctic biomes (see Fig. 3). All ordinations were performed with CANOCO version 4.0 (ter Braak and Šmilauer, 1998).

3.2. Climate reconstructions

Quantitative reconstruction of January and July air temperature, annual precipitation and July sunshine was attempted. Except July sunshine, these are the most commonly considered parameters in continental paleoclimate studies (e.g., Seppä and

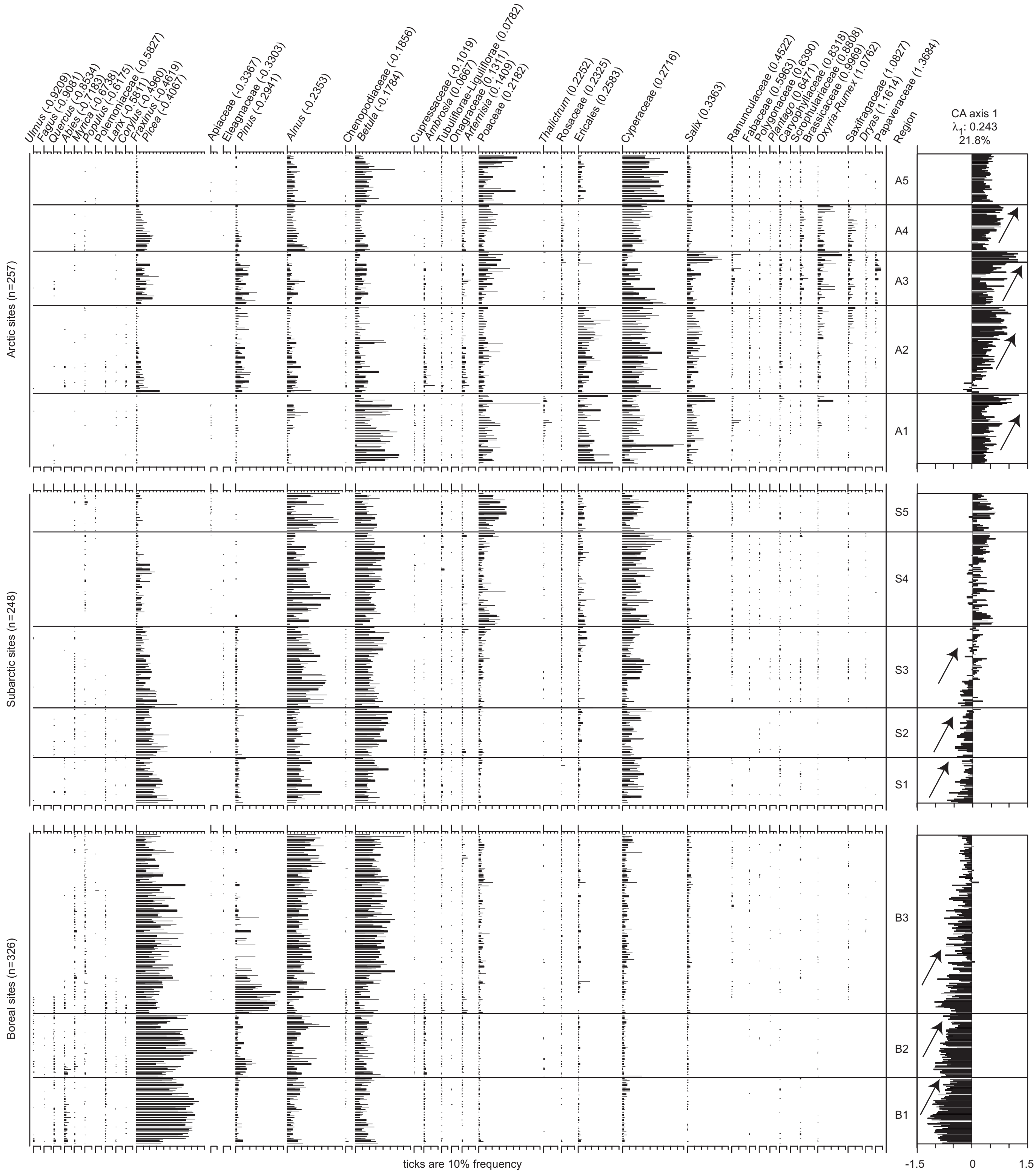

Fig. 3. The relative frequency of pollen taxa in the assemblages of the 831 modern lake sediment samples are presented by biome. There are 39 taxa pruned and re-summed from the original count data, according to procedures outlined in the text. Taxa are ordered according to their scores on the primary CCA axis constrained with July air temperature only, which appear in brackets. In general, this ordering progresses from taxa typical of warm environment (Boreal biome) on the left, to those characterizing polar sites to the right (Arctic biome). Axis 1 sample scores from correspondence analysis (CA) are given at the right. CA has been done after square-root transformation of taxa frequencies and distance joint plot scaling (site-conditional CA joint plot, i.e. scaling type 1) has been used. Arrows illustrate concomitant increases of CA axis 1 sample scores with latitude within some regions. Horizontal lines demarcate the geographic regions of Fig. 1A. Within each of these regions, assemblages are sorted according to latitude, from low (bottom) to high (top).

Birks, 2001; Davis et al., 2003; Kaufman et al., 2004; Sawada et al., 2004). Mean annual air temperature was not reconstructed because of its close relationship with January air temperature in the $n = 831$ database ($r = 0.86$) and when only arctic sites are considered ($r = 0.99$) (cf. Supplementary Table S3). Two techniques of reconstructions have been used. We focus on the modern analogue technique (MAT), but we have used CCA regressions for comparison and to evaluate the robustness of the reconstruction for a given parameter (see Guiot and de Vernal, 2007).

3.2.1. Modern analogue technique

The MAT (also known as the best modern analogue method) assumes that a fossil pollen assemblage of similar composition to a modern counterpart has been produced by similar vegetation and reflects a similar climate (Overpeck et al., 1985; Anderson et al., 1989; Guiot et al., 1989; Guiot, 1990; Andreev et al., 2003, 2004c; Jackson and Williams, 2004; Kerwin et al., 2004; Sawada et al., 2004). Similarity between fossil and modern assemblages is based on the squared chord distance (SCD) dissimilarity metric (Birks, 1977; Prentice, 1980; Overpeck et al., 1985; Gavin et al., 2003). SCD (d^2) between the fossil (t) and modern (i) assemblages is calculated from differences in relative frequencies (p), for each taxon ($j = 1$–m) ($m = 39$) as follows:

$$d_{it}^2 = \Sigma_j((p_{ij})^{1/2} - (p_{tj})^{1/2})^2$$

or

$$d_{it}^2 = 2 - 2\Sigma_j(p_{ij}p_{tj})^{1/2}.$$

The values of SCD vary between zero and 2 (the upper limit occurs when the product $p_{ij}p_{tj}$ is zero, implying that no taxon occurs simultaneously in both assemblages), the higher values indicating larger dissimilarity.

The climate parameters from fossil assemblage t (r_t) are reconstructed from the climate parameters of the best modern assemblages i (c_i) (i.e. the modern assemblages with the lowest SCD with the fossil assemblage), where i varies from 1 to s (s = number of analogues), from which the distance is d_{ti}, is given by

$$r_t = (\Sigma_i c_i/d_{ti}^2)/\Sigma_i(1/d_{ti}^2).$$

When the SCD of the closest analogues remains higher than an adopted threshold, no reconstruction can be made. The number of analogues (s) selected for interpolation of climate data is somewhat arbitrary. In general, 5–10 modern analogues are used (Andreev et al., 2003, 2004b, c, 2005; Kerwin et al., 2004; Sawada et al., 2004; Fréchette et al., 2006). Here, we used the 5 nearest analogues, which limit the risk of excessive heterogeneity within the modern analogues retained for reconstruction. All MAT reconstructions were performed with the 3Pbase software (Guiot and Goeury, 1996).

3.2.2. Threshold values of the SCD

A critical point of MAT is the definition of the threshold beyond which dissimilarity is too high for reliable analogue identification. Usually, studies comparing within-biome pollen assemblages require lower SCD threshold values than those involving more than one biome (Gavin et al., 2003; Oswald et al., 2003; Sawada et al., 2004). The SCD threshold can also be biome-specific as suggested by information-statistic indices (Waelbroeck et al., 1998) applied to continent-scale compilations of pollen assemblage (Whitmore et al., 2005).

In order to address the issue of the SCD threshold, we calculated SCDs between all pairs of the modern pollen assemblages for each biome. We then plot the frequency distribution of SCD between all pairs to illustrate dissimilarities between (1) boreal sites, (2) subarctic sites, and (3) arctic sites (Figs. 4A–C) and the frequency distribution of SCDs of the first analogue for the same three subsets of sites (Figs. 4D–F). The results demonstrate that SCDs from the Boreal biome are generally lower than those from the Arctic biome. The results for the Subarctic biome are intermediate. Furthermore, the frequency distribution of SCDs of the first analogue of sites from the Boreal biome preserves skewness and kurtosis, whereas the distribution is closer to normal for the Arctic biome. The higher modal SCD values for the Arctic biome cannot be explained by a lower density of sites (Fig. 1A). We hypothesize that higher SCD values for the Arctic biome are rather explained by higher taxonomic diversity of pollen assemblages in the Arctic biome. To test this hypothesis, we have estimated the palynological richness of modern samples by means of rarefaction analysis (Birks and Line, 1992) using the 3Pbase software (Guiot and Goeury, 1996). Rarefaction analysis aims at providing unbiased estimates of taxonomic richness for a standardized sample size (i.e., grain counts) using a random selection without replacement strategy. It produces estimates of the expected number of taxa [$E(T_n)$] for the smallest pollen sum (n) in the pollen data matrix. Rarefaction analysis was performed on the raw counts of total assemblages, including 39 spermatophytes taxa. However, we concede that the results could have been more accurate if all spermatophyte taxa (i.e. without taxonomic standardization) had been used. Nevertheless, rarefaction analyses indicate that the taxonomic richness of pollen assemblages is higher in the Arctic biome [$E(T_{102}) = 12 \pm 1$] than in the Subarctic biome (9 ± 1) and the Boreal biome (8 ± 1). We interpret the higher taxonomic richness in the Arctic biome as the result of the productivity of the local vegetation (Fréchette et al., 2006). In an open Arctic tundra, the pollen rain records influx from an association of diverse herb taxa in addition to long-distance pollen inputs. In contrast, pollen assemblages from the productive boreal forest often show lower diversity of herb pollen grains and proportionally less long-distance pollen inputs (Moore et al., 1991; Fréchette et al., 2006).

Based on these results, it seems that high SCD threshold values (>0.20) are more appropriate for MAT-based reconstructions from fossil pollen assemblages in the Arctic biome. Such thresholds reduce the probability of generating false-negative errors (i.e. Type II errors in statistical hypothesis testing), such as the omission of contrasting pollen assemblages originating from sites experiencing similar climate and therefore representing viable analogues (cf. Jackson and Williams, 2004; Wahl, 2004). Conversely, low SCD thresholds are likely to remain appropriate when considering fossil assemblages from Boreal and Subarctic biomes.

The SCD threshold as calculated in the 3Pbase programme is based on a Monte-Carlo simulation of the modern assemblage data whereby i modern spectra are randomly extracted with replacement from i selected modern spectra, and the best-fit spectrum from that group is obtained (Guiot et al., 1989; Guiot, 1990). This is repeated s times, providing s best analogues from which confidence intervals are calculated. The average SCD, generated between randomly selected pairs within the modern assemblages is 0.54 ± 0.28. The mean minus standard deviation provides the SCD threshold of 0.26. Such a SCD threshold seems adequate for quantitative paleoclimate reconstructions from pollen assemblages in Arctic Canada and Greenland because it falls in the range of SCD threshold value from Arctic biome (average SCD minus standard deviation is of 0.23; Fig. 4C).

3.2.3. CCA regression

CCA ordinations were conducted to derive regressions enabling climate reconstructions. Each CCA regression was established by constraining the 831 pollen assemblages with the climate parameters used for the reconstruction. The regression was then established by plotting the CCA axis 1 sample scores of the modern samples against the corresponding climate value. A linear

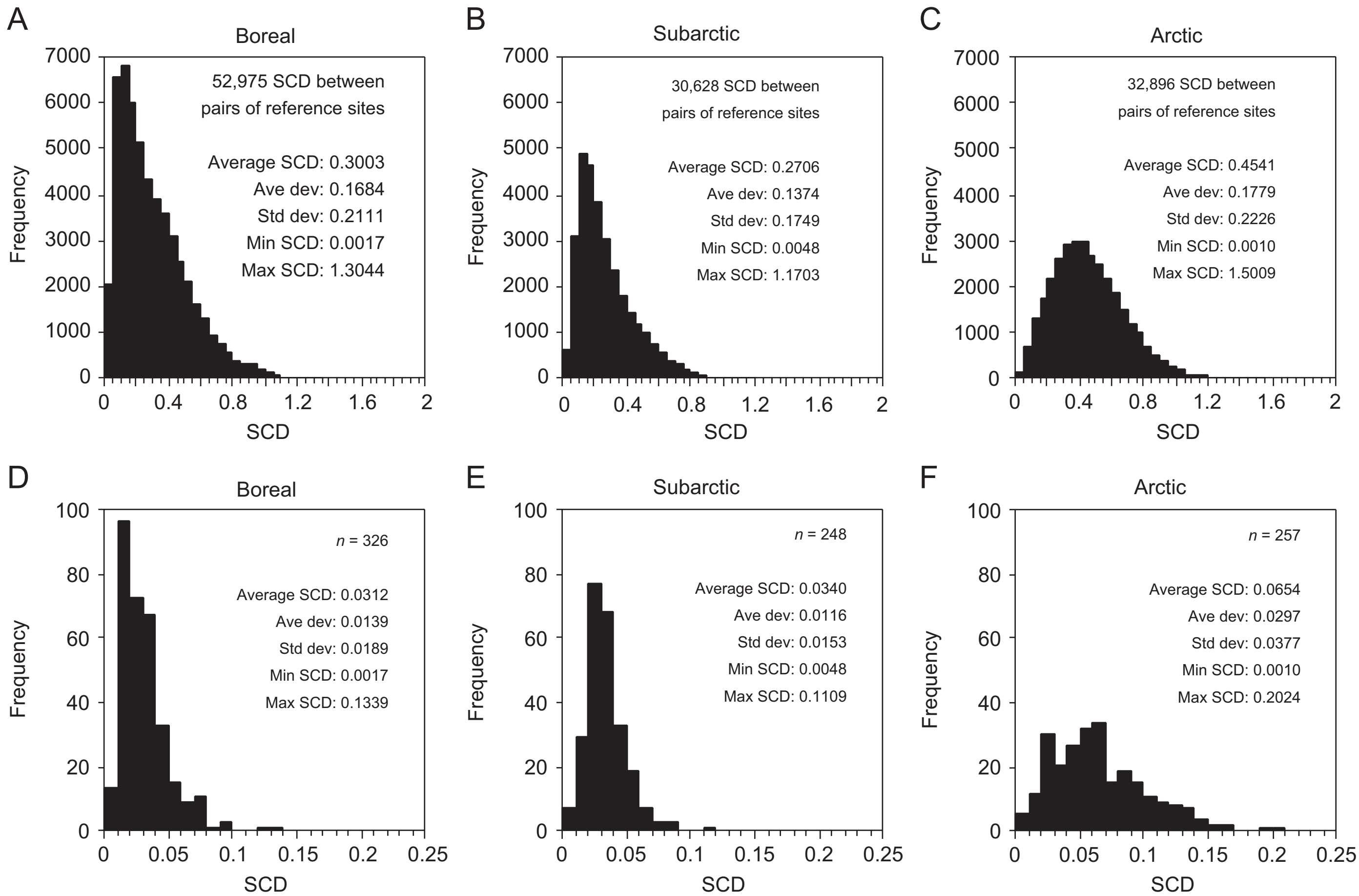

Fig. 4. Relationships between vegetation type (biome) and dissimilarities between modern pollen assemblages which is expressed as squared chord distance (SCD). The upper graphs show the frequency distributions of all pair-wise SCDs between modern pollen assemblages within the Boreal (A), Subarctic (B) and Arctic (C) biomes separately (SCD: 0.05 increments). The lower graphs show the frequency distributions of the SCD of the first (closest) analogue (MAT approach) for each modern pollen assemblages within the Boreal (D), Subarctic (E) and Arctic (F) biomes separately (SCD: 0.01 increments).

regression equation was derived for July air temperature and third-order polynomial regression equations were established for January air temperature, annual precipitation and July sunshine. Following these initial CCA ordinations of the 831 modern assemblages, fossil data were projected passively onto the ordination spaces, without influencing the analysis in any other way. The regression equations between CCA axis 1 sample scores and climate variables were then applied to fossil scores to reconstruct quantitatively past climate. Prior to ordinations, the frequencies of the 39 taxa were square-root transformed, and rare taxa were not down-weighted. All ordinations were performed with CANOCO version 4.0 (ter Braak and Šmilauer, 1998).

4. Results and discussion

4.1. Modern pollen vs. climate relationship

4.1.1. Climate variables maximizing the major floristic gradient

CA ordination reveals the major floristic gradient among the modern pollen assemblages. The first CA axis accounts for 21.8% of total variance and has an eigenvalue ($\lambda_1 = 0.243$) twice that of the second CA axis ($\lambda_2 = 0.133$) (Table 2). As illustrated in Fig. 3, the first axis shows a clear latitudinal trend in each region (cf. Fig. 1) and at the biome scale. It also shows two parallel trends that reflect east–west continental geographic differences (east and west of 90°W longitude) (Fig. 5A). The CA axis 1 is negatively correlated with July air temperature ($r = -0.73$) (Fig. 5B). Sites from the Arctic biome have positive scores on CA axis 1, whereas sites from the Boreal biome have negative scores. Sites from the Subarctic biome have both positive and negative scores. They overlap both tundra and boreal forest samples. The relationship between CA axis 1 assemblage scores and July air temperatures indicates that pollen sensitively captures the climatic distinction between boreal forest and tundra vegetation (i.e. climate of the forest tundra ecotone or treeline), which broadly coincides with the 10 °C mean July air temperature isotherm (Fig. 5B) (e.g., Bonan et al., 1992; Foley et al., 1994; Pielke and Vidale, 1995; Walker et al., 2002).

Forward-selection procedures on CCAs aim at pinpointing climate variables that maximize the floristic gradients identified by CA. The first CCA constrained the pollen assemblages to all climate variables ($n = 38$) and enabled a first-order appreciation of the climate variables that maximize the floristic variance. The variable marginal effect (*sensu* ter Braak and Verdonschot, 1995) confirms that the temperature produced the highest canonical eigenvalues among pollen assemblages from the Boreal, Subarctic and Arctic biomes collectively ($n = 831$) (Table 3). However, when each biome is considered individually the temperature does not produce the highest canonical eigenvalues in the Subarctic and Boreal biomes. In the Subarctic biome, the sunshine data maximize the floristic gradient, whereas in the boreal biome the precipitation data are determinant (Table 3). These different pollen–climate relationships are in agreement with the satellite-observed photosynthetic activity across boreal and arctic North America (Goetz et al., 2005) and the Canadian boreal forest

Table 2
Summary of CA and CCA results on the 831 modern pollen data and summary CCA results on the 3 biome subdivisions considered separately

	Axes		
	1	2	3
CA (831 sites, 39 taxa)			
Eigenvalues (λ)	0.243	0.133	0.073
% Variance explained	21.8	11.9	6.6
CCA (39 taxa, 8 environmental predictors)			
All biomes (831 sites)			
Eigenvalues (λ)	0.197	0.085	0.036
% Variance explained (taxa data)	17.7	7.6	3.2
% Variance explained (taxa–environment)	53.5	23.1	9.8
Species–environment correlation	0.902	0.811	0.750
p-value (999 Monte Carlo permutations with Bonferroni adjustment)	0.001	0.001	0.001
Sum of all unconstrained λ (total inertia): 1.114			
Sum of all canonical λ: 0.368			
Arctic biome (257 sites)			
Eigenvalues (λ)	0.124	0.050	0.043
% Variance explained (taxa data)	14.4	5.7	5.0
% Variance explained (taxa–environment)	48.5	19.4	17.0
Species–environment correlation	0.910	0.795	0.760
p-value (999 Monte Carlo permutations with Bonferroni adjustment)	0.001	0.001	0.001
Sum of all unconstrained λ (total inertia): 0.864			
Sum of all canonical λ: 0.256			
Subarctic biome (248 sites)			
Eigenvalues (λ)	0.099	0.029	0.022
% Variance explained (taxa data)	14.7	4.3	3.2
% Variance explained (taxa–environment)	51.3	15.2	11.2
Species–environment correlation	0.917	0.810	0.731
p-value (999 Monte Carlo permutations with Bonferroni adjustment)	0.001	0.001	0.001
Sum of all unconstrained λ (total inertia): 0.678			
Sum of all canonical λ: 0.194			
Boreal biome (326 sites)			
Eigenvalues (λ)	0.143	0.058	0.023
% Variance explained (taxa data)	15.6	6.4	2.5
% Variance explained (taxa–environment)	54.6	22.2	8.8
Species–environment correlation	0.928	0.843	0.75
p-value (999 Monte Carlo permutations with Bonferroni adjustment)	0.001	0.001	0.001
Sum of all unconstrained λ (total inertia): 0.916			
Sum of all canonical λ: 0.262			

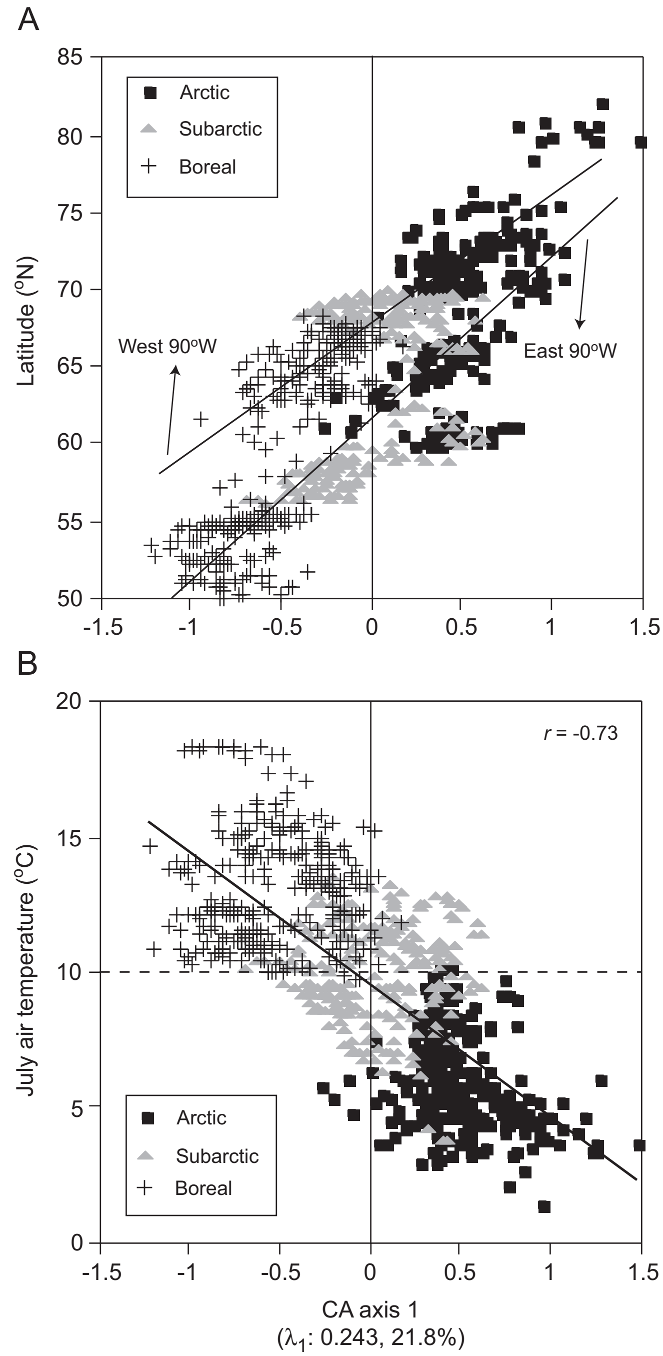

Fig. 5. Relationship between CA axis 1 sample scores and latitude (A) and July air temperature (B). Two parallel trends that reflect east–west continental geographic differences at 90°W longitude are evidenced in (A). The horizontal dashed line in (B) separates boreal forest from tundra ecotone (or treeline), which broadly coincides with the 10 °C mean July air temperature isotherm.

carbon balance (Bond-Lamberty et al., 2007), which show vegetation-specific responses to climate trends. In particular, Goetz et al. (2005) have shown that the response of tundra vegetation is more consistent with change in temperature-related variables, whereas the response of boreal forest seems more consistent with fire disturbance since precipitation exerts a more determinant effect than temperature (Bond-Lamberty et al., 2007).

At the 90% level ($p = 0.10$), the variable conditional effect (*sensu* ter Braak and Verdonschot, 1995) shows that temperature accounts for 45.5% of the variation within the overall pollen assemblages, whereas precipitation accounts for 10.2% and sunshine for 44.4% (Table 4). However, forward selections on CCAs performed on each biome individually show that temperature does not behave evenly: in the Boreal biome it accounts for 50.7% of the variation, whereas it only accounts for 7.8% and 39.5% in the Subarctic and Arctic biomes, respectively (Table 4). In the Subarctic and Arctic biomes, the sunshine explains most of the variation within the pollen assemblage data (65.1% and 47.2%, respectively).

Table 3
Marginal effects of the 38 climatic variables derived from forward selection in CCA

Variable	All sites ($n=831$)		Boreal sites ($n=326$)		Subarctic sites ($n=248$)		Arctic sites ($n=257$)	
	λ_1	% Var.	λ_1	% Var.	λ_1	% Var.	λ_1	% Var.
January T	0.046	4.1	0.043	4.7	0.032	4.7	*0.105*	*12.2*
February T	0.054	4.8	0.036	3.9	0.027	4.0	*0.102*	*11.8*
March T	0.082	7.4	0.040	4.4	0.023	3.4	*0.104*	*12.0*
April T	*0.111*	*10.0*	0.041	4.5	0.021	3.1	*0.106*	*12.3*
May T	*0.122*	*11.0*	0.026	2.8	0.030	4.4	*0.113*	*13.1*
June T	*0.120*	*10.8*	0.028	3.1	0.035	5.2	*0.087*	*10.1*
July T	*0.141*	*12.7*	0.033	3.6	0.027	4.0	0.067	7.8
August T	*0.152*	*13.6*	0.067	7.3	0.025	3.7	*0.093*	*10.8*
September T	*0.144*	*12.9*	0.078	8.5	0.023	3.4	*0.103*	*11.9*
October T	*0.118*	*10.6*	*0.119*	*13.0*	0.029	4.3	*0.097*	*11.2*
November T	0.076	6.8	*0.100*	*10.9*	0.028	4.1	*0.098*	*11.3*
December T	0.049	4.4	0.068	7.4	0.022	3.2	*0.098*	*11.3*
Annual T	0.100	9.0	0.066	7.2	0.023	3.4	*0.107*	*12.4*
Annual P	0.056	5.0	*0.099*	*10.8*	0.044	6.5	0.070	8.1
January P	0.037	3.3	0.078	8.5	0.040	5.9	0.061	7.1
February P	0.033	3.0	0.075	8.2	0.044	6.5	0.062	7.2
March P	0.045	4.0	0.081	8.8	0.039	5.8	0.064	7.4
April P	0.036	3.2	*0.094*	*10.3*	0.044	6.5	0.062	7.2
May P	0.054	4.8	*0.117*	*12.8*	0.049	7.2	0.067	7.8
June P	0.093	8.3	*0.112*	*12.2*	0.050	7.4	0.071	8.2
July P	0.108	9.7	*0.093*	*10.2*	0.045	6.6	0.080	9.3
August P	0.075	6.7	0.078	8.5	0.025	3.7	0.079	9.1
September P	0.053	4.8	*0.093*	*10.2*	0.035	5.2	0.073	8.4
October P	0.055	4.9	*0.096*	*10.5*	0.047	6.9	0.075	8.7
November P	0.048	4.3	0.091	9.9	0.050	7.4	0.074	8.6
December P	0.042	3.8	0.082	9.0	0.040	5.9	0.068	7.9
January S	0.073	6.6	0.018	2.0	0.036	5.3	0.046	5.3
February S	0.040	3.6	0.055	6.0	*0.071*	*10.5*	0.065	7.5
March S	0.054	4.8	*0.108*	*11.8*	*0.087*	*12.8*	*0.087*	*10.1*
April S	0.075	6.7	0.079	8.6	0.066	9.7	*0.103*	*11.9*
May S	0.027	2.4	0.057	6.2	*0.085*	*12.5*	0.060	6.9
June S	0.027	2.4	0.052	5.7	0.060	8.8	0.069	8.0
July S	0.037	3.3	0.059	6.4	0.034	5.0	*0.089*	*10.3*
August S	0.063	5.7	0.076	8.3	0.020	2.9	0.010	1.2
September S	0.044	3.9	0.039	4.3	*0.074*	*10.9*	*0.109*	*12.6*
October S	0.031	2.8	0.030	3.3	*0.070*	*10.3*	*0.097*	*11.2*
November S	0.036	3.2	0.062	6.8	*0.071*	*10.5*	0.031	3.6
December S	0.060	5.4	0.021	2.3	0.048	7.1	0.056	6.5

Sum of all unconstrained λ (all sites): 1.114
Sum of all unconstrained λ (Boreal sites): 0.916
Sum of all unconstrained λ (Subarctic sites): 0.678
Sum of all unconstrained λ (Arctic sites): 0.864

λ_1: fit (eigenvalue with variable *j* only).
Climate parameters that account for more than 10% of the variance within pollen data, i.e. climate parameter with the highest eigenvalues, are italicised.

4.1.2. Relationship between pollen taxa and climate variables

The second CCA constraining the pollen assemblages to eight climate variables allows the study of the relationship between pollen taxa and climate variables. The CCA results ($n=831$ sites, 39 taxa, eight climate variables) show that the first three canonical axes collectively explain 28.5% of the variance in the pollen data alone, and 86.4% of the variance in the weighted-averages of the pollen taxa constrained to climate variables (Table 2). CCA axis 1 captures the gradients of both July air temperature ($r=-0.81$) and July precipitation ($r=-0.68$), whereas axis 2 is weakly correlated with January air temperature ($r=0.47$) and July sunshine ($r=-0.47$), and axis 3 captures mainly a gradient of January air temperature ($r=0.77$) and annual precipitation ($r=0.76$) (Table 5). At the 99.9% level ($p=0.001$), all eight climate variables appear as significant predictors of the composition of pollen assemblages and were then retained for ordination biplots (Supplementary Table S2).

The CCA ordination biplots are displayed to show climatic gradients as well as the positions of the 39 taxa and the 831 sites on the first two ordination axes (Figs. 6A and B). The separation of pollen assemblages among biomes is maximized along axis 1, which corresponds to the July air temperature gradient as also shown by CA. By projecting taxon scores onto the vector of July air temperature, three distinct groupings are observed. This ranking of taxa approximates the weighted-averages of individual pollen taxa with respect to July air temperature (Jongman et al., 1987; Legendre and Legendre, 1998). The first group is characterized by strongly negative CCA axis 1 scores and includes the most thermophilous and most abundant taxa in the Boreal biome. The second group with strongly positive scores includes the most common taxa in the Arctic biome. The last group embraces taxa that are generally more abundant in the Subarctic biome, but that are also encountered in samples from Boreal to High Arctic biomes (Fig. 3). This group therefore has somewhat lower ecological specificity with respect to the parameterized climatic gradients. Axis 2 is more strongly influenced by January air temperature and July sunshine, which are almost opposite to each other especially in the Arctic biome ($r=-0.71$) (Supplementary Table S3).

The position of pollen taxa in relation to climatic gradients is also displayed for each biome individually (Figs. 6C, D, and F). For the Arctic biome, taxa of the coldest environments (High Arctic) have negative scores on axis 1 whereas those encountered in Low

Table 4
Conditional effects of the 38 climatic variables derived from forward selection in CCA

Variable	All sites ($n=831$)			Boreal sites ($n=326$)			Subarctic sites ($n=248$)			Arctic sites ($n=257$)		
	λ_a	% Var.	p-value	λ_a	% Var.	p-value	λ_a	% Var.	p-value	λ_a	% Var.	p-value
January T	0.012	2.3	0.001	0.002	0.5	0.363	0.006	2.0	0.001	0.003	0.7	0.099
February T	0.002	0.4	0.003	0.003	0.8	0.097	0.002	0.7	0.193	0.006	1.4	0.001
March T	0.003	0.6	0.001	0.010	2.6	0.001	0.002	0.7	0.479	0.003	0.7	0.138
April T	0.017	3.2	0.001	0.004	1.0	0.014	0.005	1.7	0.001	0.008	1.9	0.001
May T	0.007	1.3	0.001	0.003	0.8	0.091	0.002	0.7	0.673	*0.113*	*27.3*	*0.001*
June T	0.003	0.6	0.001	0.004	1.0	0.008	0.004	1.3	0.001	0.006	1.4	0.001
July T	0.001	0.2	0.017	0.002	0.5	0.223	0.002	0.7	0.676	0.004	1.0	0.011
August T	*0.152*	*28.5*	*0.001*	0.002	0.5	0.133	0.002	0.7	0.674	0.002	0.5	0.272
September T	0.001	0.2	0.009	0.014	3.6	0.001	0.004	1.3	0.002	0.006	1.4	0.001
October T	0.014	2.6	0.001	*0.119*	*30.4*	*0.001*	0.002	0.7	0.23	0.003	0.7	0.105
November T	0.029	5.4	0.001	0.003	0.8	0.067	0.003	1.0	0.094	0.005	1.2	0.003
December T	0.001	0.2	0.002	0.034	8.7	0.001	0.002	0.7	0.579	0.003	0.7	0.173
Annual T	0.001	0.2	0.211	0.002	0.5	0.345	0.002	0.7	0.679	0.004	1.0	0.004
Annual P	0.002	0.4	0.001	0.003	0.8	0.086	0.006	2.0	0.001	0.005	1.2	0.002
January P	0.004	0.8	0.001	0.003	0.8	0.027	0.015	5.0	0.001	0.004	1.0	0.012
February P	0.007	1.3	0.001	*0.053*	*13.5*	*0.001*	0.006	2.0	0.001	0.003	0.7	0.212
March P	0.003	0.6	0.001	0.003	0.8	0.04	0.004	1.3	0.004	0.003	0.7	0.052
April P	–	–	–	0.003	0.8	0.03	0.003	1.0	0.085	0.004	1.0	0.006
May P	0.004	0.8	0.001	0.010	2.6	0.001	0.003	1.0	0.086	0.010	2.4	0.001
June P	0.003	0.6	0.001	0.008	2.0	0.001	0.014	4.7	0.001	0.011	2.7	0.001
July P	0.003	0.6	0.001	0.005	1.3	0.005	0.002	0.7	0.356	0.002	0.5	0.805
August P	0.004	0.8	0.001	0.003	0.8	0.08	0.003	1.0	0.009	0.003	0.7	0.188
September P	0.009	1.7	0.001	0.003	0.8	0.021	0.003	1.0	0.023	–	–	–
October P	0.005	0.9	0.001	0.001	0.3	0.964	–	–	–	0.006	1.4	0.001
November P	0.004	0.8	0.001	0.003	0.8	0.099	0.003	1.0	0.036	0.004	1.0	0.002
December P	0.006	1.1	0.001	–	–	–	0.016	5.3	0.001	0.005	1.2	0.001
January S	0.031	5.8	0.001	0.003	0.8	0.054	0.007	2.3	0.001	0.006	1.4	0.001
February S	0.002	0.4	0.001	0.006	1.5	0.001	0.002	0.7	0.341	*0.045*	*10.9*	*0.001*
March S	0.044	8.3	0.001	0.014	3.6	0.001	*0.087*	*28.9*	*0.001*	0.003	0.7	0.109
April S	0.019	3.6	0.001	0.004	1.0	0.003	0.022	7.3	0.001	0.009	2.2	0.001
May S	0.006	1.1	0.001	0.025	6.4	0.001	0.003	1.0	0.052	0.041	9.9	0.001
June S	0.007	1.3	0.001	0.004	1.0	0.011	0.011	3.7	0.001	0.014	3.4	0.001
July S	0.046	8.6	0.001	0.003	0.8	0.072	0.003	1.0	0.007	0.003	0.7	0.083
August S	0.004	0.8	0.001	0.004	1.0	0.014	0.003	1.0	0.063	0.014	3.4	0.001
September S	*0.067*	*12.6*	*0.001*	0.004	1.0	0.004	*0.034*	*11.3*	*0.001*	0.033	8.0	0.001
October S	0.003	0.6	0.001	0.012	3.1	0.001	0.003	1.0	0.025	0.003	0.7	0.026
November S	0.003	0.6	0.001	0.005	1.3	0.001	0.006	2.0	0.001	0.003	0.7	0.052
December S	0.004	0.8	0.001	0.008	2.0	0.001	0.004	1.3	0.001	0.014	3.4	0.001
Sum of all constrained λ (all sites): 0.533												
Sum of all constrained λ (Boreal sites): 0.414												
Sum of all constrained λ (Subarctic sites): 0.301												
Sum of all constrained λ (Arctic sites): 0.392												

λ_a: additional fit (increase in eigenvalue).
p-value: significance level of the effect obtained from Monte Carlo permutations under the reduced model with 999 random permutations.
Climate parameters that account for more than 10% of the variance within pollen assemblage data, i.e. climate parameter with the highest eigenvalues, are italicised.

Arctic have positive scores. The boreal component (*Picea* and *Pinus*) of the pollen assemblages has also negative scores on axis 1 as the high Arctic component. In High Arctic environments, long-distance pollen inputs are often over-represented in the assemblages because of low productivity of local plants (Bourgeois et al., 2001; Gajewski, 2002). For example, in the Canadian Arctic Archipelago (see region A3 in Fig. 1A), *Picea* and *Pinus* account for 18% of the pollen assemblages in average (Fig. 3).

4.2. Robustness of quantitative climate reconstructions

Validation tests were made to evaluate the accuracy of quantitative reconstructions of January and July air temperature, July sunshine and annual precipitation (Fig. 7). Ideally, the reconstructions of modern conditions should yield a 1:1 linear relationship between inferred and observed values. The accuracy of reconstruction can be assessed from the slope and the correlation coefficient (r), as well as from the standard deviation of the difference between inferred and observed values that corresponds to the root mean square error (RMSE) (e.g., de Vernal et al., 2001; Sawada et al., 2004).

4.2.1. Modern analogue technique

Validation tests on modern samples demonstrate the ability of the MAT to reconstruct modern climate parameters in the Boreal, Subarctic and Arctic biomes with a reasonable accuracy for lake-sediment localities (Figs. 7A and B). For each climate variable, the reconstruction error (RMSE) is close to their interpolation error (RTGCV, see Section 2.2). For North America and Greenland (20–85°N, 20–180°W), the RTGCV of the gridded July and January air temperature is estimated to be 0.7 and 1.1 °C, respectively, and the RTGCV of the gridded July sunshine is estimated to be 4.5% (New et al., 1999, 2002). Correlation coefficients suggest that July and January air temperatures can be reconstructed with confidence ($r=0.92$ and 0.89, respectively), as well as precipitation and July sunshine ($r=0.91$ and 0.90, respectively). The accuracy of July air temperature predictions is better in the Arctic biome relative to Subarctic and Boreal biomes (RMSE of 1.12 °C versus

Table 5
Intra-set correlation matrix between CCA axes 1–3 and the 8 climate variables

Variable	Environment axis 1	Environment axis 2	Environment axis 3
All biomes (831 sites)			
January *T*	−0.139	0.473	0.773
July *T*	−0.807	0.328	−0.101
Annual *T*	−0.604	0.420	0.581
January *P*	−0.272	0.089	0.652
July *P*	−0.684	0.000	0.605
Annual *P*	−0.400	0.035	0.755
January *S*	−0.521	0.334	−0.319
July *S*	−0.079	−0.471	−0.464
Arctic biome (257 sites)			
January *T*	0.909	0.083	−0.019
July *T*	0.693	−0.214	0.261
Annual *T*	0.917	−0.034	−0.044
January *P*	0.663	0.164	−0.102
July *P*	0.764	0.017	−0.284
Annual *P*	0.714	0.150	−0.193
January *S*	−0.246	−0.844	0.119
July *S*	−0.815	−0.191	0.272
Subarctic biome (248 sites)			
January *T*	0.366	−0.653	0.049
July *T*	0.299	−0.415	0.624
Annual *T*	0.118	−0.727	0.129
January *P*	−0.524	−0.387	0.315
July *P*	−0.562	−0.607	−0.001
Annual *P*	−0.550	−0.572	0.099
January *S*	0.508	−0.401	−0.020
July *S*	0.335	0.808	0.275
Boreal biome (326 sites)			
January *T*	0.359	0.507	0.034
July *T*	0.264	−0.503	0.472
Annual *T*	0.627	0.134	0.130
January *P*	0.531	0.800	0.121
July *P*	0.718	0.547	−0.171
Annual *P*	0.704	0.686	−0.125
January *S*	−0.063	0.091	−0.048
July *S*	0.346	−0.798	0.415

Results are given for the entire database and for the Boreal, Subarctic and Arctic biomes independently.

1.20 and 1.23 °C, respectively) (Table 6). Conversely, the other climate parameters are more accurately reconstructed south of the Arctic biome. The precipitation is best reconstructed in the Boreal and Subarctic biomes and the sunshine is best reconstructed in the Subarctic biome. These results are in agreement with the forward-selection results of the climate variables within CCAs (Tables 3 and 4).

4.2.2. *Evaluating the potential effects of spatial autocorrelation in MAT reconstructions*

MAT is one of the most commonly used technique in paleoclimatology (cf. Guiot and de Vernal, 2007) but is subject to criticisms by Telford et al. (2004) and Telford and Birks (2005) who claim that the reliability of the MAT is overestimated because of spatial autocorrelation that may distort the performance of certain calculations, such as that of standard statistical tests (*r*-value and RMSE) (Legendre, 1993). In order to address this issue, we made tests by removing sites from the modern reference database (*N*). We first sorted the sites by latitude and we subdivided the data set by retaining three sites over four ($N = 623$), then one out of two ($N = 416$) and finally only one out of four ($N = 208$). The four most significant climate parameters (July and January air temperature, July sunshine, annual precipitation) were estimated from MAT using the different data subsets. In the last iteration ($N = 208$), the closeness, geographically and climatologically, among sites comprising the training set is reduced, as is the risk for statistical artefacts due to spatial autocorrelation. Comparable RMSE results were found for the $N = 208$, 416, 623 and 831 reference samples (Table 7). We therefore conclude that spatial autocorrelation does not affect the RMSE in MAT and does not constitute a limitation of the approach. On the contrary, the validation tests presented here show that high-latitude pollen assemblages can be used to reconstruct past climates (cf. also Gajewski, 2002), notably the July air temperature and sunshine. Paleoclimate reconstructions using MAT are methodologically and statistically robust.

4.2.3. *CCA regression*

CCA regressions produce quantitative climate reconstructions with a much lower accuracy than MAT as shown in Figs. 7C and D. As with MAT, the best results are obtained with July air temperature. The results also show that the reconstruction of annual precipitation should be interpreted with caution, especially at sites experiencing abundant precipitations such as the Greenland coasts.

4.3. *Application to short sequences*

Fossil pollen assemblages from the top 4 cm of two lacustrine sedimentary sequences were selected as examples. The first site is from eastern Cumberland Peninsula on Baffin Island (Akvaqiak Lake; 66°47′N, 63°57′W, 45 m a.s.l.) and the second one is from southwest Greenland (Qipisarqo Lake; 61°00′N, 47°45′W, 7 m a.s.l.) (Fig. 1B). The respective location of Akvaqiak and Qipisarqo lakes is suitable to document changes in synoptic scale atmospheric circulation patterns since they lie in the trajectory of one of the principal North Atlantic cyclone tracks (cf. Tsukernik et al., 2007 and references therein).

Both lakes are from low elevations, in a protected valley or lowland at the extremity of fjords. The cores were collected in 1998 with a modified K-B gravity core (Glew, 1989). The Akvaqiak Lake sediments studied here encompass the last 450 years (radiocarbon dates of 650 cal years BP at 6.25 cm), whereas the sediments analysed at Qipisarqo Lake only cover the 20th century (Kaplan et al., 2002). Pollen concentrations in Qipisarqo Lake sediments are about half those observed in Akvaqiak Lake, which can be explained by higher sediment accumulation rate at the Greenland site (Fig. 8).

The climate conditions at both sites show important contrasts, especially in winter. The present-day July and January air temperature average, respectively, 4.9 and −24.1 °C on eastern Baffin Island (Environment Canada, 2004) and 8.4 and −6.2 °C in southwest Greenland (Cappelen et al., 2001). Annual precipitation is about 360 mm on eastern Baffin Island and 860 mm in southwest Greenland. The modern sea-ice cover (with >50% of sea-ice concentration) averages 8–9 months/year along the eastern coast of Baffin Island whereas it averages only 1–2 months/year along southwest Greenland (from the 1953–2000 data set provided by the National Snow and Ice Data Center (NSIDC) in Boulder, CO, USA). The climatic contrast observed on both sides of the Davis Strait—Labrador Sea region is related to regional ocean circulation patterns.

Despite contrasting climate, the local vegetation of these sites is comparable. Vegetation is moderately lush shrub tundra dominated by shrub birch (*Betula*) and heaths (Ericales). This similarity is well expressed in their respective pollen assemblages (Fig. 8). The SCD between the uppermost pollen assemblage of Akvaqiak and Qipisarqo lacustrine sequences is 0.14, which indicates that they are indeed very similar to each other (cf. Overpeck et al., 1985; Anderson et al., 1989).

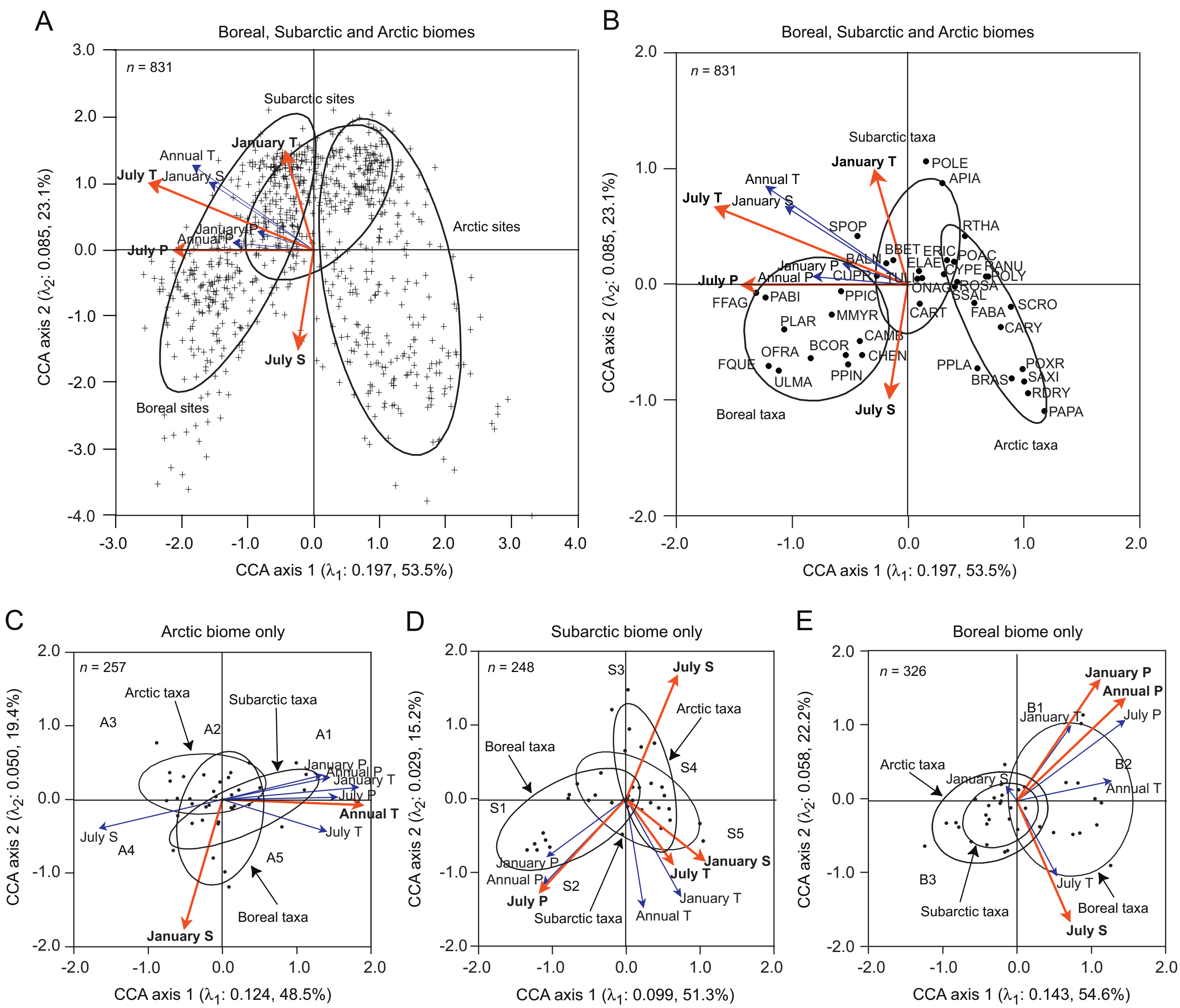

Fig. 6. Canonical correspondence analysis (CCA) ordination biplots of modern pollen assemblages using correlation biplot scaling (taxa-conditional biplot, i.e. scaling type 2) for which distances between pollen taxa approximate their interspecific correlation. Red arrows indicate forward selection (Supplementary Table S2). Graph A: Biplots representing all sites ($n = 831$) and climatic variables (arrows) ($n = 8$). 95% ellipses encircle each biome. Graph B: Biplots representing taxa ($n = 39$) and climatic variables (arrows) ($n = 8$). Taxon codes are given in Table 1. 95% ellipses encircle taxa mainly encountered in the Boreal, Subarctic and Arctic biomes. Graphs C–E. Biplots representing taxa ($n = 39$) and climate variables (arrows) ($n = 8$) within the Arctic biome ($n = 257$ sites), the Subarctic biome ($n = 248$), and the Boreal biome ($n = 326$). 95% ellipses include same taxa as in graph B. In graphs C–E, the approximate location of sites grouped by region is indicated. For clarity, the length of all arrows have been multiplied by 3 (A) or 2 (B–E). Note that January air temperature and July sunshine gradients have opposite direction in graphs A and C.

4.3.1. Comparing surface sediment reconstruction with observed value

Quantitative reconstruction of climate parameters at Akvaqiak and Qipisarqo lakes are illustrated in Fig. 9, with reference to MAT and CCA regressions. Close analogues for the Akvaqiak and Qipisarqo lakes pollen assemblages are identified in the modern database, with at least five analogues having SCD <0.14, which is much lower than the threshold of 0.26 (see Section 3.2.2).

The comparison of reconstructed climate from the uppermost pollen assemblage from both sequences with observed climate data may be used to ascertain that the developed methodology is reliable to reconstruct past climates from Arctic pollen assemblages (Table 8). The observed climate data at Akvaqiak and Qipisarqo lakes were estimated via lapse-rate collected bilinear interpolation, identically to the 831 modern sites (see Whitmore et al., 2005 for the method). Although pollen assemblages preserved in sediment reflect longer time interval than the instrumental climate data, we may consider that an approach failed if the reconstructed value falls outside the range of the prediction error (RMSE) of the climate value or if the range of the reconstructed value (maximum minus minimum value) is too large when using MAT.

The MAT approach succeeded to reconstruct July air temperature and July sunshine from the uppermost sample at both sites (Table 8). It succeeded to reconstruct January air temperature and annual precipitation at the Greenland site, but yielded overestimated values at the Baffin Island site. The CCA approach succeeded to reconstruct all variables at the Baffin Island site and climate trends are generally comparable with those from MAT. At the Greenland site, only July air temperature was correctly reconstructed with CCA regressions. However, the failure of the

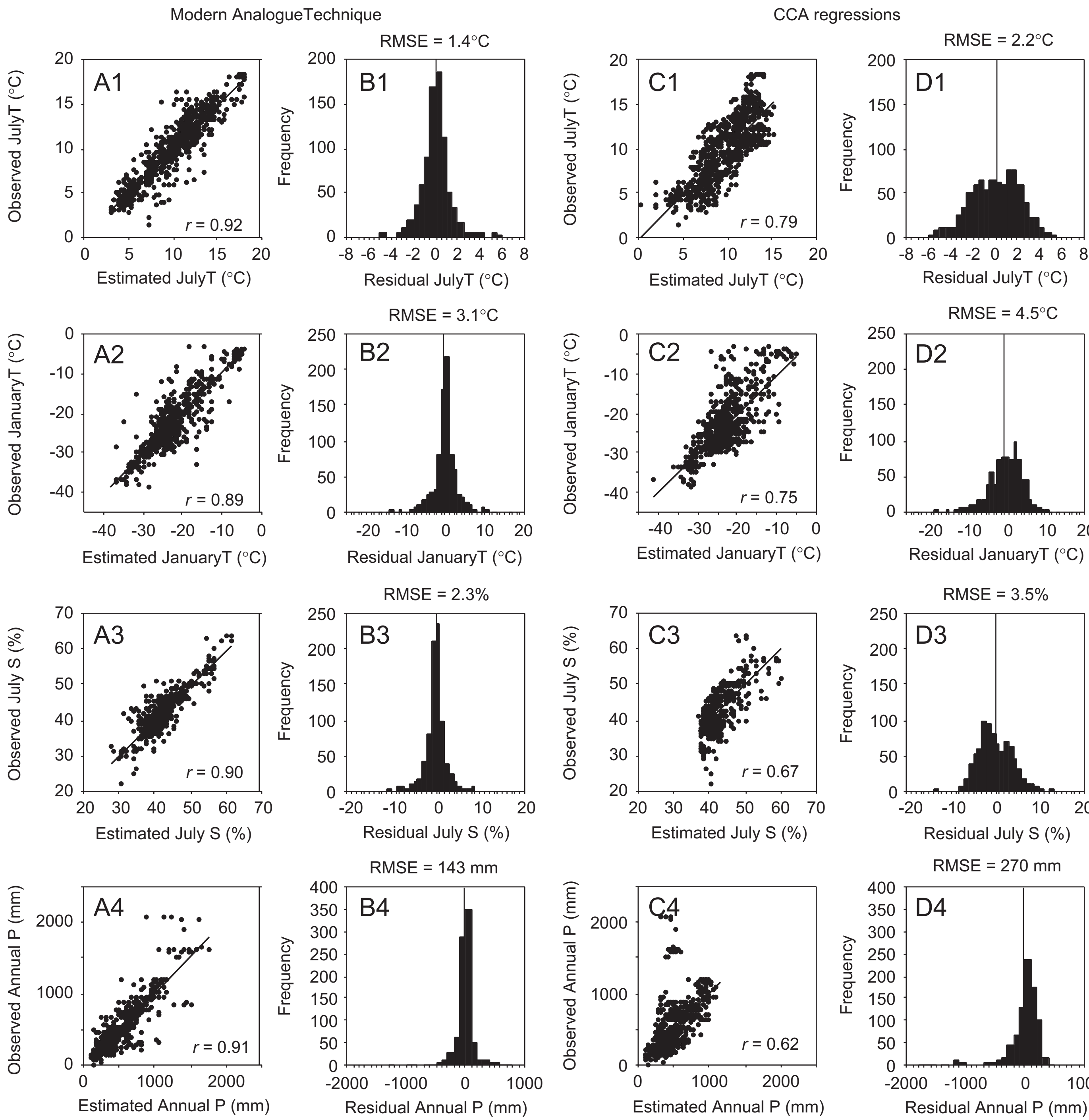

Fig. 7. Comparison between measured and pollen-inferred modern climate parameters for the 831 sites (A, C), and distribution of residuals (estimated−observed values) (B, D) from the MAT and the CCA regressions approaches. In each case, linear correlation coefficients (r) and root mean squared errors (RMSE) provide indication on the accuracy of estimates. Each CCA regressions was established by constraining the 831 pollen assemblages with the climate variable at the reconstruction.

Table 6
Summary results of validation tests using the modern analogue technique with the entire database ($n = 831$) and for the Boreal, Subarctic and Arctic biomes considered separately

Sites	January T (R)	January T RMSE (°C)	July T (r)	July T RMSE (°C)	July S (r)	July S RMSE (%)	Annual P (R)	Annual P RMSE (mm)
All sites ($n = 831$)	0.89	3.10	0.92	1.35	0.90	2.3	0.91	143
Arctic sites ($n = 257$)	0.92	3.76	0.78	1.12	0.90	2.5	0.89	206
Subarctic sites ($n = 248$)	0.88	2.56	0.74	1.20	0.87	1.7	0.88	98
Boreal sites ($n = 326$)	0.71	2.95	0.81	1.23	0.91	2.5	0.93	110

Table 7
Results of validation tests with MAT made on different subsets of data to address the question of spatial autocorrelation and its potential effect on the error of prediction

Parameter	r	RMSE
First configuration: N = 831, n = 831		
January *T*	0.89	3.10 °C
July *T*	0.92	1.35 °C
Annual *P*	0.91	143 mm
July *S*	0.90	2.34%
Number of sites with less than 5 analogues: 4		
Mean SCD of the first analogue: 0.042		
Mean SCD of the fifth analogue: 0.067		
Second configuration: N = 623, n = 208		
January *T*	0.87	3.26 °C
July *T*	0.92	1.38 °C
Annual *P*	0.90	145 mm
July *S*	0.89	2.33%
Number of sites with less than 5 analogues: 4		
Mean SCD of the first analogue: 0.043		
Mean SCD of the fifth analogue: 0.070		
Third configuration: N = 416, n = 415		
January *T*	0.88	3.21 °C
July *T*	0.93	1.33 °C
Annual *P*	0.91	145 mm
July *S*	0.91	2.21%
Number of sites with less than 5 analogues: 4		
Mean SCD of the first analogue: 0.046		
Mean SCD of the fifth analogue: 0.080		
Fourth configuration: N = 208, n = 623		
January *T*	0.82	3.94 °C
July *T*	0.92	1.41 °C
Annual *P*	0.82	201 mm
July *S*	0.87	2.68%
Number of sites with less than 5 analogues: 15		
Mean SCD of the first analogue: 0.056		
Mean SCD of the fifth analogue: 0.100		

The 12 estimates are based on the 5 best analogues with 4 configurations: 831 reference sites (N) producing 831 inferences (n), 623 reference sites (N) producing 208 inferences (n), 416 reference sites (N) producing 415 inferences (n) and 208 reference sites (N) yielding 623 inferences (n).

CCA method at the Greenland site is not surprising given the poor accuracy obtained with validation tests (see Section 4.3.2).

4.3.2. *Trends of last 500 years*

A compilation of paleoclimate records based on varve and dark lamination thicknesses, tree ring widths, glaciers summer melt and marine proxies revealed that the circum-Arctic warmed approximately by 1.5 °C from the nineteenth to the twentieth centuries (Overpeck et al., 1997). The paleoclimate archives of Overpeck et al. (1997) also evidenced a spatial asymmetry of climate and environmental changes over the last 400 years. Kaufman et al. (2004) and Kerwin et al. (2004) documented the non-uniform trend of Holocene Arctic climate.

At both Greenland and Baffin Island sites, the reconstructed July air temperature from MAT and CCA regression approaches closely match each other (Fig. 9). The MAT- and CCA-based reconstructions of July sunshine also closely match each other at Akvaqiak Lake but not at Qipisarqo Lake. Notable discrepancies between MAT- and CCA-based reconstructions are obtained for January air temperature and annual precipitation at both sites. On these bases, variations in July air temperature can be inferred with high confidence, whereas the January air temperature and annual precipitation should be considered cautiously.

At Akvaqiak Lake, where the sequence spans a longer time interval of about 450 years, no trends are evidenced for any of reconstructed parameters nor a clear recent summer temperature warming (Fig. 9). This contrasts with the MAT-based result at Qipisarqo Lake, which shows a 20th century increase of July and January air temperatures together with a decrease of July sunshine, or increase of cloud cover, consistent with an increase in annual precipitation.

Nevertheless, during the 20th century, a cold episode was recorded at Akvaqiak Lake (Fig. 10B). A decrease of about 0.5 °C in July air temperature is observed. Based on a calibrated date of 650 years BP (or 1350 AD) at 6.25 cm and on uniform sediment accumulation rate between 1350 AD and today (i.e. 1998 AD), we estimated that the cooling pulse occurred between 1900 and 1970 AD. A similar summer (JJA) cold event is reconstructed in record from varved sediments at Donard Lake (Moore et al., 2001), in the Cape Dyer region about 95 km east of Akvaqiak Lake (Fig. 10A). At Donard Lake, the cooling is centred around 1930 AD (1910–1960 AD). Over the Arctic, a mid-20th century cooling (1940–1970) of annual surface air temperature (SAT) of about 0.75 °C has been reported by Moritz et al. (2002). A cold SAT period between 1940 and 1980 AD has also been recorded in borehole temperature profiles in northern Quebec, Canada (Chouinard et al., 2007). In northern Quebec, the cooling in SAT was of ca 0.4 K. We thus suggest that the cooling episode around 1930 AD observed on eastern Baffin Island corresponds to the cooling signal observed by Moritz et al. (2002) over the Arctic, and by Chouinard et al. (2007) in northern Quebec. The slight asynchronism observed among records could result from dating uncertainties of lacustrine sediments.

The recent summer warming at Qipisarqo Lake is subtle in comparison to the 20th century warming reported by Overpeck et al. (1997). This could result from the delay in the response of the vegetation, which is birches-dominated shrub tundra, to the 20th century climate warming. Proxies from both terrestrial (pollen for example) and aquatic (varve thicknesses, diatoms, chironomids, notably) ecosystems react in different ways to the same climatic and environmental changes due to differences in controlling factors, time lags and other feedback mechanisms (Douglas et al., 1994; Abbott et al., 2000; Rosén et al., 2001; Bigler et al., 2002). Wagner et al. (2000) have proposed that the more rapid response of the lake environment to climatic shifts is due to the reduced lifetime of freshwater organisms, often combined with a high reproduction rate.

The increasing trend of cloud cover at Qipisarqo Lake over the last 100 years suggests an increase in cyclone frequency, which might be related to changes in synoptic scale atmospheric circulation patterns. In southwestern Greenland, increase in cyclone frequency and cloudiness are associated to negative NAO index (Cappelen et al., 2001; Previdi and Veron, 2007). Therefore, our results suggest a progressive shift from a dominant positive NAO at the beginning of the 20th century to a negative NAO onward. Serreze et al. (2000) showed that summer cyclone frequency and annual precipitation north of 60°N increased from the mid-20th century to present (Figs. 11A and B), and Hurrell (1995) showed that the July NAO index shifted from a dominant positive mode to a dominant negative one from ca 1920 AD onward (Fig. 11C) (NAO Index Data provided by the Climate Analysis Section, NCAR, Boulder, USA (Hurrell, 1995), see http://www.cgd.ucar.edu/cas/jhurrell/indices.html for updated NAO time series).

4.4. *Strengths and weaknesses of MAT to reconstruct arctic climate*

On the whole, the validation tests (Table 6 and Fig. 7) and the climate reconstructions performed from sequences of eastern Baffin Island and southern Greenland (Table 8 and Fig. 9) demonstrate the robustness of MAT, especially with regard to the July air temperature and sunshine. The reconstructed modern

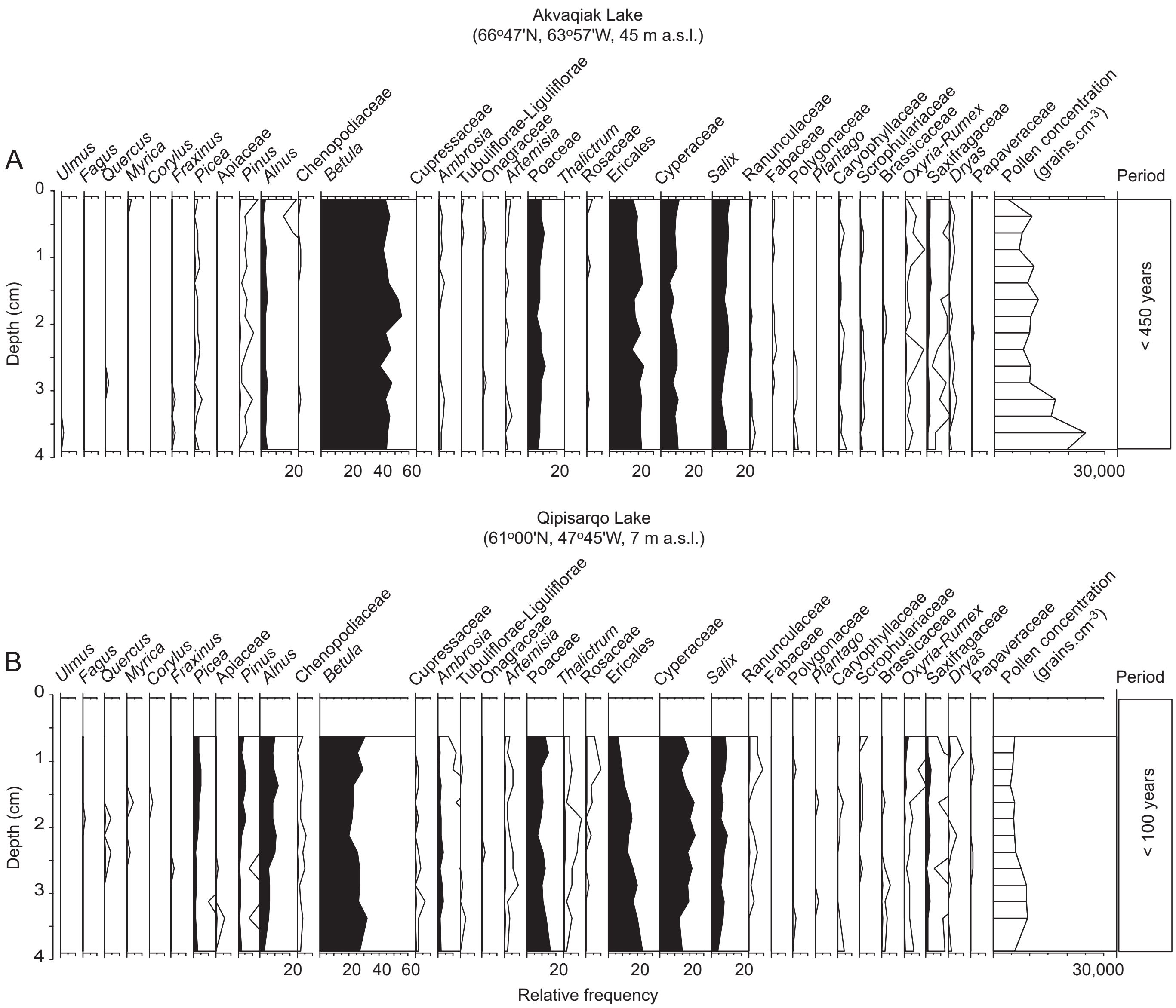

Fig. 8. Summary pollen diagrams from Akvaqiak (A) and Qipisarqo (B) lakes. Unfilled curves correspond to 10 × exaggerations of relative frequencies. Pollen sums and the ordering of taxa are as in Fig. 3 (warmer on left, colder on right). To allow a better between-core comparison we used the same upper limit scales for each taxa.

values of these two climate parameters fall within the range of their respective RMSE (±1.4 °C for July air temperature and ±2.3% for July sunshine). They also show that the two parameters, July sunshine and July air temperature, can be reconstructed simultaneously and yield independent trends (the correlation (r) between modern July sunshine and modern July air temperature is of 0.30, see Supplementary Table S3). However, the exercise revealed that the quantitative reconstruction of annual precipitation and January air temperature should be made with caution.

The lower accuracy of estimated annual precipitation can in part be related to the low precision of the gridded modern precipitation data because of the inherent spatial variability of this climate parameter (New et al., 1999, 2002). However, it also comes from the selected analogues. At Akvaqiak Lake, January air temperature and annual precipitation are strongly overestimated because three of the five best modern analogues selected for climate calculation of the uppermost sample come from south Greenland, where annual precipitation and January temperature are relatively high (Cappelen et al., 2001), thus leading to overestimation. Given the SCD ranging from 0.05 to 0.08, the three Greenland modern samples are very close analogues (cf. Overpeck et al., 1985; Anderson et al., 1989), but they are possibly misleading because they present very similar pollen assemblages despite different annual precipitation and winter temperature (i.e. false analogues, analogous to Type I error in statistical hypothesis statistic). The similarity between the topmost pollen assemblages of Akvaqiak and Qipisarqo lakes (SCD of 0.14) further illustrate this false analogy.

In order to discard false analogues, Ortu et al. (2006) have constrained the selection of analogues by a biomization procedure (Prentice et al., 1996). During the selection of the analogues, the biome assigned to the fossil assemblage is compared to the modern samples and modern analogues for which the biomes are not consistent are rejected (Guiot et al., 1993). The method was successful for Italian mountainous pollen sites (Ortu et al., 2006), but such a procedure is not helpful here because the vegetation type (or biome) surrounding Akvaqiak and Qipisarqo lakes is similar, i.e. an erect drawf-shrub tundra (*sensu* BIOME 4 Model of Kaplan et al., 2003).

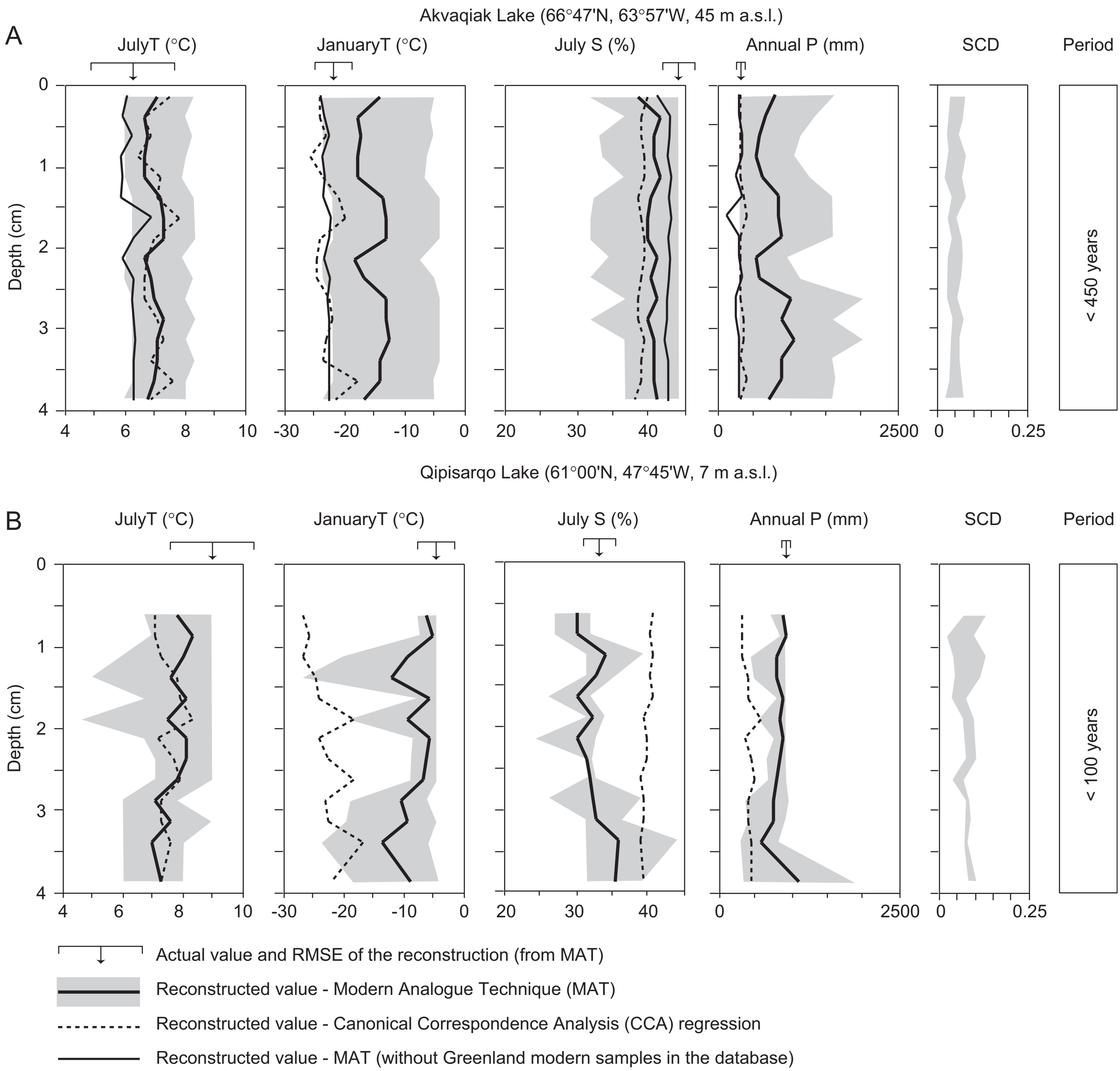

Fig. 9. Climate reconstruction from pollen assemblages of Akvaqiak (A) and Qipisarqo (B) lake cores based on the modern analogue technique (MAT) and canonical correspondence analysis (CCA) regressions. Modern value of all climate parameters is illustrated by an arrow on the top of each graph. The RMSE (see Fig. 7B) of the climate parameters is also illustrated. The thick black lines represent reconstructed value from MAT, the thin black lines represent reconstructed value from MAT after exclusion of Greenland samples in the modern database and the dashed lines represent reconstructed value from CCA regressions. The grey envelope is the confidence intervals for each parameter corresponding to the five best modern analogues with Greenland samples included in the modern database. The SCD of the best analogues is given on the right of the diagram. To allow a better between-core comparison we used the same upper limit scales for each climate parameter.

To reconstruct more adequately all climate parameters at Akvaqiak Lake for the recent interval, Greenland samples could be discarded from the modern database (Table 8 and Fig. 9). Such a procedure is based on a geographical constraint as proposed by Pflaumann et al. (1996) for the reconstruction of sea-surface temperature based on planktic foraminifera. This procedure may resolve apparent biases due to the automatic selection of false analogues from distant regions. The procedure is probably adequate for recent intervals, but it cannot be justified for the reconstruction of older intervals corresponding to *a priori* unknown analogues.

This raises important questions about the pollen vs. climate relationships. Are pollen assemblages in the Arctic capturing the signal of winter air temperature and annual precipitation ? Is the Greenland vegetation located at the edge of a glaciated Island in "equilibrium" with climate conditions ? Part of the response comes from multivariate analyses (Section 4.1) showing that July air temperature and sunshine are the most determinant parameters on pollen assemblages in the Arctic and Subarctic biomes, whereas annual precipitation and January temperatures are more weakly related.

Table 8
Comparison of observed and reconstructed climate values for the uppermost sediment sample at Akvaqiak Lake (Baffin Island) and Qipisarqo Lake (Greenland)

Variable	Observed value		Reconstructed value (MAT)				Reconstructed value (CCA)		
	Baffin	Greenland	RMSE	Baffin	Baffin*	Greenland	RMSE	Baffin	Greenland
January T (°C)	−22.0	−4.4	3.1	−14.2	−24.2	−6.0	4.5	−24.0	−26.8
Range (°C)				−23.6 to −5.0	−32.9 to −19.4	−8.0 to −4.4			
July T (°C)	6.2	9.0	1.4	7.1	6.0	7.9	2.2	7.5	7.1
Range (°C)				6.0–8.2	4.1–9.6	6.8–9.0			
July S (%)	44.3	33.3	2.3	38.6	41.1	30.2	3.5	40.0	40.8
Range (%)				31.7–44.3	37.2–44.3	26.9–31.8			
Annual P (mm)	299	948	143	794	304	873	270	313	301
Range (mm)				299–1630	232–372	724–948			
SCD				0.03–0.08	0.03–0.14	0.06–0.13			

Baffin: Akvaqiak Lake.
Baffin*: Akvaqiak Lake without modern Greenland samples ($n = 59$) in the modern pollen database.
Greenland: Qipisarqo Lake.
Range: Reconstructed value of the first and fifth analogue.
SCD: SCD of the first and fifth analogue.

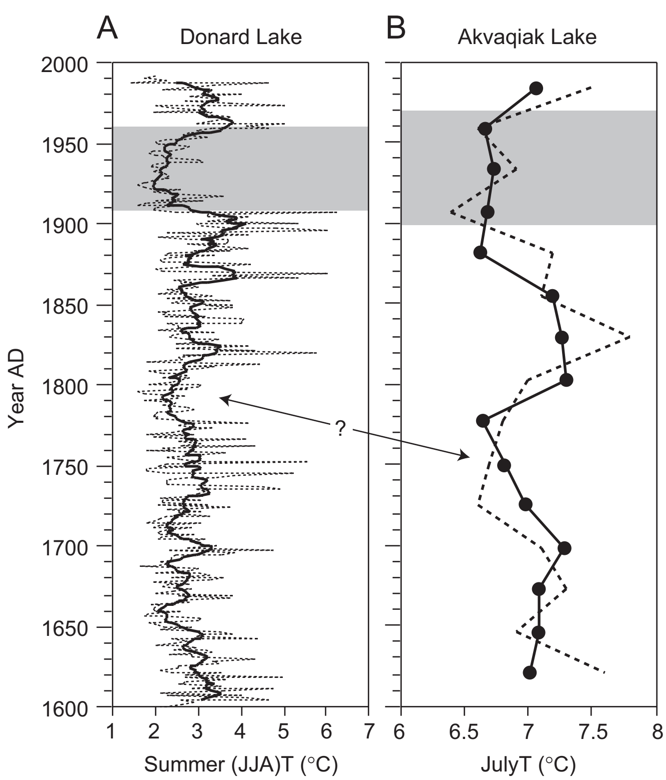

Fig. 10. Comparison between July air temperature reconstructed at Akvaqiak Lake and summer (JJA) temperature estimated at Donard Lake (Moore et al., 2001) for the last 400 years. (A) Summer temperature record for Donard Lake calculated from varve thickness (Moore et al., 2001). The dashed line represents reconstructed value for each year from 1600 AD to 1992 AD and the thick line represents a 10-year running average to emphasize decadal trends and reduce noise from the annual record. (B) July air temperature record for Akvaqiak Lake. The solid line represents reconstructed value from MAT and the dashed line represents reconstructed value from CCA regressions.

5. Conclusion

The present study of 831 modern pollen assemblages from surface sediments collected in lakes from the Boreal, Subarctic and Arctic biomes of North America and Greenland allowed us to document the pollen vs. climate relationships taking into account several parameters that include monthly temperature, precipitation and sunshine data, average annual temperature and total annual precipitation. Multivariate analyses (CA and CCA) demonstrate that climate information related to these parameters is well recorded in pollen assemblages. However, they also demonstrate different weight for each of these parameters depending upon the biome analysed. Our results show that temperature parameters are best captured by pollen assemblages of the Boreal and Arctic biomes, whereas sunshine is best registered in pollen assemblages from the Subarctic and Arctic biomes, and precipitation is best registered in pollen assemblages from the Boreal and Subarctic biomes. This is in agreement with studies of ecophysiological changes resulting from global climate change showing that temperature-related variables induce change in Arctic biome whereas change in Boreal biome seems to have been driven by fire frequency and precipitation (Goetz et al., 2005; Bond-Lamberty et al., 2007). The variable pollen vs. climate relationship from one vegetation type to another should thus be taken into account when developing models or applying transfer functions to reconstruct past climate from arctic, subarctic or boreal pollen assemblages.

Two approaches have been tested for quantitative climate reconstruction: the MAT and CCA regressions. Validation tests show that CCA regressions yield on the whole inaccurate results, which tends to indicate that the pollen response to climate is non-linear, nor simply unimodal. On the contrary, MAT, which does not rely on calibration, yields more accurate estimates of climate parameters. However, in the Arctic biome, the mean error of prediction is relatively high for annual precipitation and January air temperature while July air temperature and sunshine are well reconstructed. The bias toward overestimated annual precipitation and winter temperature raises a number of research questions concerning the pollen–climate relationship. It underlines the need for more adequate strategies when using modern pollen samples for quantitative reconstruction of annual precipitation and January air temperature in high-latitude sites. One strategy could be the use of a constraint based on other paleoclimatic proxies (Guiot et al., 1993).

The applications of MAT to recent sediment samples from Akvaqiak Lake on eastern Baffin Island, and Qipisarqo Lake on southwest Greenland demonstrated that summer temperature and sunshine can be reconstructed from arctic pollen assemblages. The comparison of their trends with other summer climate reconstructions from independent proxies supports their reliability.

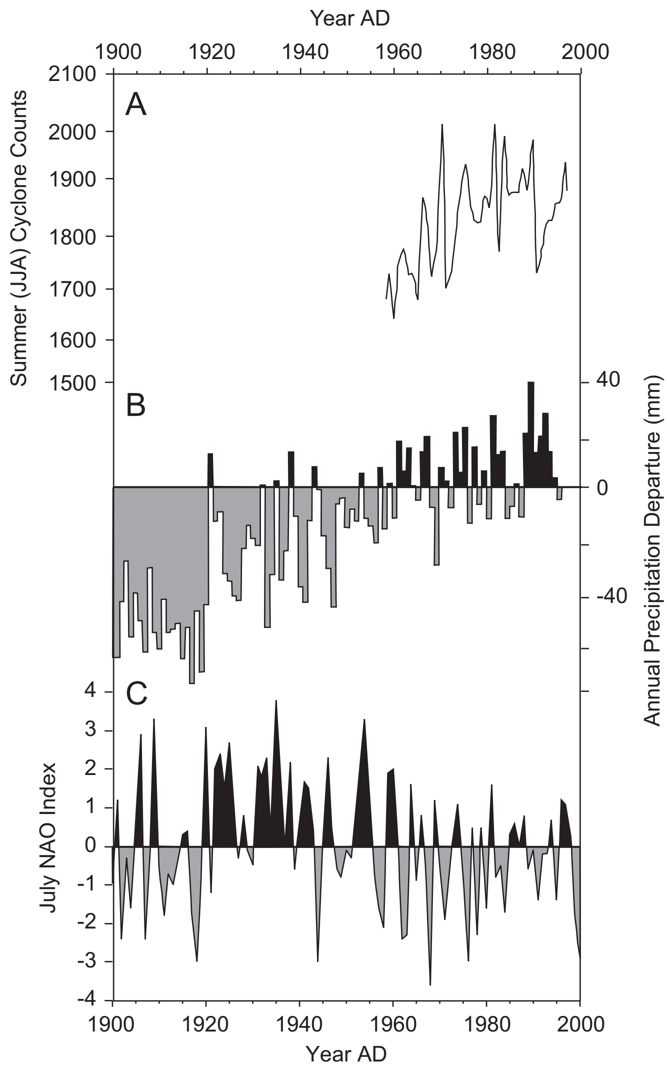

Fig. 11. (A) Time series of summer (JJA) cyclone counts north of 60°N for 1958–1997 AD (Serreze et al., 2000). (B) Annual precipitation anomalies (55–85°N) for 1900–1995 AD with respect to 1951–1980 AD means (mm) (Serreze et al., 2000). (C) July index of the North Atlantic Oscillation (NAO) based on the difference of normalized sea level pressures between Ponta Delgada, Azores and Stykkisholmur/Reykjavik, Iceland since 1900 (NAO Index Data provided by the Climate Analysis Section, NCAR, Boulder, USA, Hurrell, 1995). The updated time series is available at http://www.cgd.ucar.edu/cas/jhurrell/indices.html.

The reconstructions indicate relatively uniform conditions over the last centuries at the eastern Baffin site with, however, a mid-20th century summer cooling period consistent with other paleoclimate records (Moore et al., 2001; Moritz et al., 2002; Chouinard et al., 2007). In contrast, a 20th century warming trend accompanied by increased cloud cover (i.e., decreased sunshine) is evidenced at the southern Greenland site. The increase in nebulosity over the last century in southern Greenland is consistent with increasing summer cyclone frequency and annual precipitation over high-latitudes from the mid-20th century onward (Serreze et al., 2000), and suggest a shift in the July NAO index from a dominant positive mode at ca 1920 AD to a dominant negative one at ca 1990 AD (Hurrell, 1995).

Despite low and highly variable pollen production of tundra plants, the present study demonstrates that pollen assemblages permit climate reconstructions in arctic environment (see also Gajewski, 2006), notably with respect to summer air temperature and sunshine parameters. The reconstruction of past trends of these two climate parameters is of great interest for arctic paleoclimate research, including synoptic scale atmospheric circulation patterns like the NAO.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.quascirev.2008.02.016.

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